Measure-to-measure interpolation using Transformers

A machine-checked blueprint: the original statements and proofs, our Lean formalization, and what it strengthens

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About this blueprint

This is a standalone, A-to-Z blueprint of the controllability results of

> M. Geshkovski, P. Rigollet, Y. Ruiz-Balet, *Measure-to-measure interpolation using Transformers*, arXiv:2411.04551v3,

together with their machine-checked Lean 4 formalization. It is meant to be read on its own: for each node it reproduces the **original statement (and proof, where short) verbatim**, then gives **our formalization** (a faithful statement in Lean, a prose proof, the real source listing, and an honest status badge), and, where relevant, **what our work strengthens** (a corrected sign error, a discharged rigor gap, or exactly what is axiomatised).

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How to read the boxes.

> **From the original paper.** Grey boxes reproduce the authors' statements and proofs verbatim, with a page-cited attribution.

> **Our formalization.** Blue boxes hold our Lean statement, a prose proof, the real source listing, and a status badge: machine-checked (clean Lean kernel proof), axiomatised (rests on a labeled axiom standing in for missing Mathlib theory), open (stated in Lean as a `sorry`), or informal (reviewed on paper).

> **What we strengthen.** Green boxes record where the formalization corrects or completes the paper: a sign error (F1), a rigor gap discharged by a kernel-checked leaf (F2), or what had to be axiomatised.

The status badge of every machine-checked node is verified against the Lean kernel by `#print axioms` (`bin/axiom-report`): a node is badged machine-checked only if its axiom set is clean (`propext`, `Classical.choice`, `Quot.sound` only).

1 Foundations

The paper views a Transformer as the flow map of a continuity equation on the unit sphere $\mathbb{S}^{d-1}$. We reproduce the setting verbatim, then give the kernel-checked algebraic foundations our leaves rest on.

From the original paper (Geshkovski, Rigollet, Ruiz-Balet, §1, pp. 2-3).
Viewing each token as a particle, given an initial sequence $(x_1(0), \dots, x_n(0)) \in (\mathbb{S}^{d-1})^n$, one considers

$$\dot{x}_i(t) = \mathsf{v}[\mu(t)](t, x_i(t)), \qquad t \in [0, T], \tag{1.1}$$

where $\mu(t) = \frac{1}{n}\sum_{j=1}^{n} \delta_{x_j(t)}$ is the empirical measure. The vector field is

$$\mathsf{v}[\mu](t, x) = \mathbf{P}_x^{\perp}(\mathbf{V}(t)\, \mathcal{A}_{\mathbf{B}}[\mu](t, x) + \mathbf{W}(t)(\mathbf{U}(t)x + b(t))_+), \tag{1.2}$$

where the self-attention is

$$\mathcal{A}_{\mathbf{B}}[\mu](t, x) := \int e^{\langle \mathbf{B}(t)x, x' \rangle} x' \, \mu(\mathrm{d}x') \Big/ \int e^{\langle \mathbf{B}(t)x, \zeta \rangle} \mu(\mathrm{d}\zeta).$$

The vector field is “ultimately projected onto $\mathsf{T}_x\mathbb{S}^{d-1}$ by virtue of the orthogonal projector $\mathbf{P}_x^{\perp} := I_d - xx^{\top}$, referred to as *layer normalization*.” Since (1.1) depends only on $\mu(t)$, one passes to the **continuity equation**

$$\partial_t \mu(t) + \mathrm{div}(\mu(t)\, \mathsf{v}[\mu(t)]) = 0 \;\text{ on } [0, T] \times \mathbb{S}^{d-1}, \qquad \mu(0) = \mu_0. \tag{1.3}$$

For any control $\theta = (\mathbf{V}, \mathbf{B}, \mathbf{W}, \mathbf{U}, b)$ the Cauchy problem (1.3) is well-posed and yields a continuous, invertible flow map $\Phi_\theta^t : \mathcal{P}(\mathbb{S}^{d-1}) \to \mathcal{P}(\mathbb{S}^{d-1})$ with $\Phi_\theta^t(\mu_0) = \mu(t)$.

1.1 Tangential projector (layer normalization)

Our formalization. machine-checked
We define the layer-normalization projector $\mathbf{P}_x^{\perp} v = v - \langle x, v \rangle x$ and prove the algebraic identities every leaf uses (ledger node **L1**): it is linear, symmetric, idempotent on the sphere, annihilates x, and satisfies the quadratic identity

$$\langle \mathbf{P}_x^{\perp} v, v \rangle = \|v\|^2 - \langle x, v \rangle^2 \qquad (\|x\| = 1),$$

which is exactly the scalar driving the gate ODE (B.5) and the barycenter ODE (B.9). The proof is a direct computation: expanding $\langle v - \langle x, v \rangle x,\ v \rangle$ and using $\langle x, x \rangle = 1$ cancels the cross term to $\|v\|^2 - \langle x, v \rangle^2$.

```
/-- The tangential projector `P_x^⊥ v = v - ⟪x, v⟫ x`. -/
noncomputable def tangentialProjector (x v : Eucl d) : Eucl d := v - (⟪x, v⟫) • x

/-- The key quadratic identity (L1): for a unit vector `x`,
`⟪P_x^⊥ v, v⟫ = ‖v‖² - ⟪x, v⟫²`. -/
theorem projector_inner_sub_sq {x : Eucl d} (hx : x ∈ sphere d) (v : Eucl d) :
    ⟪tangentialProjector x v, v⟫ = ‖v‖ ^ 2 - ⟪x, v⟫ ^ 2 := by
  have h : ⟪x, x⟫ = 1 := inner_self_eq_one_of_mem_sphere hx
  simp only [tangentialProjector, inner_sub_left, inner_smul_left, RCLike.conj_to_real,
    real_inner_self_eq_norm_sq]
  ring
```

Foundations/Projector.lean

1.2 Geodesic distance

Our formalization. machine-checked

The geodesic distance on the sphere is $d_g(x, y) = \arccos\langle x, y\rangle$. We prove the inversion $\cos d_g(x, y) = \langle x, y\rangle$ for unit vectors and that $d_g(x, y) \in [0, \pi]$. These let the gate sign (B.4) be read either as the inner-product inequality $\langle z, x\rangle < \cos R$ or as the geodesic statement $d_g(z, x) > R$ (via strict antitonicity of cos on $[0, \pi]$).

```
/-- Geodesic distance on the unit sphere: `d_g(x, y) = arccos⟪x, y⟫`. -/
noncomputable def geodesicDist (x y : Eucl d) : ℝ := Real.arccos (⟪x, y⟫)

/-- `cos (d_g x y) = ⟪x, y⟫`: the cosine of the geodesic distance is the inner product.
 ↪ -/
theorem cos_geodesicDist {x y : Eucl d} (hx : x ∈ sphere d) (hy : y ∈ sphere d) :
    Real.cos (geodesicDist x y) = ⟪x, y⟫ :=
  Real.cos_arccos (neg_one_le_inner hx hy) (inner_le_one hx hy)
```

Foundations/GeodesicDistance.lean

2 Self-contained leaves

These are the self-contained computational cores the paper's proofs rest on, each isolated from its measure-theoretic and flow-map scaffolding and given a clean Lean kernel proof. Two of them (L7, L8) reduce instead to a single labeled Wasserstein axiom and are flagged as such; the gate leaf (L2) and the non-colinearity leaves (L11, L11′) additionally repair or discharge a gap in the original text.

2.1 Gate algebra and gate ODE

From the original paper (Geshkovski, Rigollet, Ruiz-Balet, Lemma B.2 and eq. (B.4)-(B.5), pp. 31-32).
Lemma B.2 transports mass between two overlapping balls $\mathcal{B}_0, \mathcal{B}_1 \subset \mathbb{S}^{d-1}$ ($\mathcal{B}_0 \cap \mathcal{B}_1 \neq \emptyset$): for any $\varepsilon > 0$, $T > 0$ there exist $W, U \in \mathcal{M}_{d\times d}(\mathbb{R})$ and $b \in \mathbb{R}^d$ so that the solution μ satisfies $\mu(T, \mathcal{B}_0 \cap \mathcal{B}_1) \geq (1-\varepsilon)\mu_0(\mathcal{B}_0)$. In the proof:
"Let $z \in \mathbb{S}^{d-1}$ denote the center and $R > 0$ the radius of $\mathcal{B}_0$. Take an arbitrary $\omega \in \mathrm{int}(\mathcal{B}_0 \cap \mathcal{B}_1)$. We consider $U = -\mathbf{1}z^\top$ and $b = \cos(R)\mathbf{1}$, as well as any $W \in \mathcal{M}_{d\times d}(\mathbb{R})$ such that $W\mathbf{1} = \omega$. Then $W(Ux+b)_+ = (-\cos d_g(z,x) + \cos(R))_+\,\omega$, and note that"

$$(-\cos d_g(z,x) + \cos(R))_+ > 0 \quad\Longleftrightarrow\quad x \in \mathcal{B}_0. \tag{B.4}$$

"Now observe that"

$$\frac{\mathrm{d}}{\mathrm{d}t}\langle x(t), \omega\rangle = (-\cos d_g(z, x(t)) + \cos(R))_+(1 - \langle x(t), \omega\rangle^2), \tag{B.5}$$

"which is positive whenever $x(t) \in \mathcal{B}_0 \setminus \{\omega\}$."

Our formalization. machine-checked
The gated drift is $W(Ux+b)_+ = g\cdot\omega$ with scalar gate $g = (\cos R - \langle z, x\rangle)_+$. Projecting onto the tangent space and pairing with the unit vector ω, the projector identity L1 collapses the tangential part to a scalar: $\langle \mathbf{P}_x^\perp(g\,\omega), \omega\rangle = g\,\langle \mathbf{P}_x^\perp\omega, \omega\rangle = g(\|\omega\|^2 - \langle x,\omega\rangle^2) = g(1-\langle x,\omega\rangle^2)$ since $\|\omega\| = 1$ — this is `gate_inner_identity`. The ODE (B.5) is then the chain rule: $\frac{\mathrm{d}}{\mathrm{d}t}\langle x(t),\omega\rangle = \langle x'(t), \omega\rangle$ (ω constant), and substituting the projected drift $x'(t) = \mathbf{P}_{x(t)}^\perp(g\,\omega)$ reproduces the logistic factor. For the sign (B.4), $0 < \cos R - \langle z, x\rangle \iff \langle z, x\rangle < \cos R$, and rewriting $\langle z, x\rangle = \cos d_g(z,x)$ with strict antitonicity of cos on $[0,\pi]$ gives the geodesic form. The key step is the quadratic projector identity, which turns the ambient projection into the scalar $1 - \langle x, \omega\rangle^2$.

```
/-- L2 (algebra of B.5): for unit `x` and unit `ω`, projecting the scaled drift `g•ω`
 ↪  and pairing
with `ω` gives the logistic factor `g(1 - ⟪x,ω⟫²)`. -/
theorem gate_inner_identity {x ω : Eucl d} (hx : x ∈ sphere d) (hω : ω ∈ sphere d) (g :
 ↪  ℝ) :
    ⟪tangentialProjector x (g • ω), ω⟫ = g * (1 - ⟪x, ω⟫ ^ 2) := by
  rw [tangentialProjector_smul, real_inner_smul_left, projector_inner_sub_sq hx ω,
    norm_eq_one_of_mem_sphere hω, one_pow]

/-- L2 (the gate ODE B.5): along a path `x(t)` on the sphere whose velocity is the
 ↪  projected gated
drift `x'(t) = P_{x(t)}^⊥ (g • ω)`, the inner product `⟪x(t), ω⟫` evolves as
`g(1 - ⟪x(t), ω⟫²)`. -/
theorem gate_hasDerivAt_inner {x : ℝ → Eucl d} {ω : Eucl d} {t : ℝ} {x' : Eucl d}
    (hx : HasDerivAt x x' t) (hxs : x t ∈ sphere d) (hω : ω ∈ sphere d) (g : ℝ)
    (hode : x' = tangentialProjector (x t) (g • ω)) :
    HasDerivAt (fun s ⇒ (⟪x s, ω⟫ : ℝ)) (g * (1 - ⟪x t, ω⟫ ^ 2)) t := by
  have hconst : HasDerivAt (fun _ : ℝ ⇒ ω) (0 : Eucl d) t := hasDerivAt_const t ω
  have h := hx.inner ℝ hconst
  rw [hode, gate_inner_identity hxs hω g] at h
  simpa using h

/-- L2 (gate sign, B.4, geodesic form): the gate is active exactly outside the closed
 ↪  cap of radius
`R` around `z`, i.e. on `{x : d_g(z, x) > R}`. -/
theorem gate_pos_iff_dist {z x : Eucl d} (hz : z ∈ sphere d) (hx : x ∈ sphere d)
    {R : ℝ} (hR : R ∈ Set.Icc (0 : ℝ) Real.pi) :
    (⟪z, x⟫ : ℝ) < Real.cos R ↔ R < geodesicDist z x := by
  rw [← cos_geodesicDist hz hx]
  exact Real.strictAntiOn_cos.lt_iff_gt (geodesicDist_mem_Icc z x) hR
```

Leaves/GateODE.lean

Strengthening (finding F1). The printed (B.4) asserts the gate is active iff $x \in \mathcal{B}_0 = B(z, R)$. Our kernel-checked `gate_pos_iff_dist` proves instead $(\cos R - \langle z, x\rangle)_+ > 0 \iff d_g(z, x) > R$ — the gate is active on the **complement** of the cap. The opposite sign is the consistent one: (B.5) needs the gate active *inside* $\mathcal{B}_0 \setminus \{\omega\}$ to push interior mass toward ω, whereas the printed parameters $U = -\mathbf{1}z^\top$, $b = \cos(R)\mathbf{1}$ leave the interior mass on a zero gate, so the lemma as written transports nothing; the fix is $U = +\mathbf{1}z^\top$, $b = -\cos(R)\mathbf{1}$. This is corroborated independently by experiment E1 (the gated-transport mass fraction rises from ≈ 0.27 to 1.0 once the seed is placed on $\{d_g(z, x) > R\}$) and by the paper's own §4 Step 3, which states the identical construction with the correct side. See `ERRATA.md`.

2.2 Barycenter ODE

From the original paper (Geshkovski, Rigollet, Ruiz-Balet, eq. (B.9), p. 33). In Step 1 of the proof of Lemma 3.3, the solution to

$$\begin{cases} \partial_t \mu(t) + \mathrm{div}\big(\mathbf{P}_x^{\perp} \langle \alpha_1, \mathbb{E}_{\mu(t)}[x] \rangle \, \alpha_1 \, \mu(t)\big) = 0 & \text{on } \mathbb{R}_{\geq 0} \times \mathbb{S}^{d-1}, \\ \mu(0) = \mu_0 & \text{on } \mathbb{S}^{d-1} \end{cases} \tag{B.9}$$

for $\mu_0 \in \mathcal{P}(\mathbb{Q}_1^{d-1})$ satisfies

$$\frac{\mathrm{d}}{\mathrm{d}t} \langle \mathbb{E}_{\mu(t)}[x], \alpha_1 \rangle = \langle \mathbb{E}_{\mu(t)}[x], \alpha_1 \rangle \Big(1 - \int \langle x', \alpha_1 \rangle^2 \mu(t, \mathrm{d}x')\Big).$$

Our formalization. machine-checked
The disentangling field is $\mathbf{P}_x^{\perp}(c\,\alpha)$ with $c = \langle \mathbb{E}_\mu[x], \alpha \rangle$, so the per-particle characteristic $\langle x(t), \alpha \rangle$ obeys exactly the gate identity of L2 with c in place of the gate scalar: $\frac{\mathrm{d}}{\mathrm{d}t} \langle x(t), \alpha \rangle = c\,(1 - \langle x(t), \alpha \rangle^2)$. The Lean derivative is therefore literally `gate_hasDerivAt_inner` instantiated at $g = c$. The sign lemma `barycenter_deriv_pos` records that when $c > 0$ and the component lies strictly inside $(-1, 1)$ the right-hand side is positive, the dissipation that drives $\langle x, \alpha \rangle \to 1$, i.e. $x \to \alpha$. The key point is that the paper shows c is sign-preserved along the flow, so the logistic ODE is monotone.

```
/-- L6 (derivative, eq. B.9): along `x'(t) = P_{x(t)}^⊥ (c • α)` with `α` a unit vector,
↪ the
component `⟪x(t), α⟫` evolves as `c(1 - ⟪x(t), α⟫²)`. -/
theorem barycenter_hasDerivAt_inner {x : ℝ → Eucl d} {α : Eucl d} {t : ℝ} {x' : Eucl d}
    (hx : HasDerivAt x x' t) (hxs : x t ∈ sphere d) (hα : α ∈ sphere d) (c : ℝ)
    (hode : x' = tangentialProjector (x t) (c • α)) :
    HasDerivAt (fun s ↦ (⟪x s, α⟫ : ℝ)) (c * (1 - ⟪x t, α⟫ ^ 2)) t :=
  gate_hasDerivAt_inner hx hxs hα c hode

/-- L6 (sign / strict increase): when the barycenter coefficient `c` is positive and the
↪ component
is strictly inside `(-1, 1)`, the component strictly increases. This is the dissipation
↪ that pushes
`⟪x, α⟫` toward `1`. -/
theorem barycenter_deriv_pos {c y : ℝ} (hc : 0 < c) (hy : y ∈ Set.Ioo (-1 : ℝ) 1) :
    0 < c * (1 - y ^ 2) := by
  obtain ⟨h1, h2⟩ := hy
  have hy2 : y ^ 2 < 1 := by nlinarith
  have : 0 < 1 - y ^ 2 := by linarith
  positivity
```

Leaves/BarycenterODE.lean

2.3 Geodesic-distance gradient

From the original paper (Geshkovski, Rigollet, Ruiz-Balet, eq. after (4.4), p. 20). "Observe that the trajectories of the ODE"

$$\dot{x}(t) = (\langle \gamma, x(t)\rangle - \epsilon/2)_+ \, \mathbf{P}^{\perp}_{x(t)}\omega_+ \qquad \text{for } t \geq 0, \tag{4.4}$$

"follow the Riemannian gradient flow of the distance between ω_+ and x in $\mathcal{S}_+$. Indeed,"

$$\nabla_1 d_g(x, \omega_+) = -\frac{\mathbf{P}^{\perp}_x(\omega_+)}{\sqrt{1 - \langle x, \omega_+\rangle^2}}.$$

Our formalization. machine-checked

Two facts pin down the gradient. First, the chain rule for $\arccos \circ \langle \cdot, \omega\rangle$: since $(\arccos)'(u) = -1/\sqrt{1-u^2}$ (Mathlib's `Real.hasDerivAt_arccos`), $\frac{\mathrm{d}}{\mathrm{d}t} d_g(x(t), \omega) = -(1 - \langle x, \omega\rangle^2)^{-1/2}\langle x', \omega\rangle$. Second, for a tangent velocity ($\langle x', x\rangle = 0$) the ambient ω may be replaced by its tangential projection: $\langle x', \mathbf{P}^{\perp}_x\omega\rangle = \langle x', \omega\rangle - \langle x, \omega\rangle\langle x', x\rangle = \langle x', \omega\rangle$. Hence the derivative is $\langle x', -(1 - \langle x, \omega\rangle^2)^{-1/2}\,\mathbf{P}^{\perp}_x\omega\rangle$, exhibiting $-\mathbf{P}^{\perp}_x\omega/\sqrt{1-\langle x,\omega\rangle^2}$ as the Riemannian (tangential) gradient. The key step is that tangency annihilates the cross term, letting the Euclidean inner product against ω be read as the inner product against the gradient direction.

```
/-- L4 (derivative of geodesic distance along a path): the chain rule for `arccos∘⟪·,
 ↪ ω⟫`. -/
theorem geodesicDist_hasDerivAt {x : ℝ → Eucl d} {ω : Eucl d} {t : ℝ} {x' : Eucl d}
    (hx : HasDerivAt x x' t) (h1 : (⟪x t, ω⟫ : ℝ) ≠ -1) (h2 : (⟪x t, ω⟫ : ℝ) ≠ 1) :
    HasDerivAt (fun s => geodesicDist (x s) ω)
      (-(1 / Real.sqrt (1 - (⟪x t, ω⟫ : ℝ) ^ 2)) * ⟪x', ω⟫) t := by
  have hinner : HasDerivAt (fun s => (⟪x s, ω⟫ : ℝ)) (⟪x', ω⟫) t := by
    have h := hx.inner ℝ (hasDerivAt_const t ω)
    simpa using h
  have h := (Real.hasDerivAt_arccos h1 h2).comp t hinner
  simpa only [geodesicDist, Function.comp_def] using h

/-- L4 (gradient direction): on the sphere, for a tangent velocity the inner product
 ↪ with `ω` equals
the inner product with the tangential projection `P_x^⊥ ω`. Hence the geodesic-distance
 ↪ derivative
is `⟪x', -(1/√(1-⟪x,ω⟫²)) • P_x^⊥ ω⟫`, exhibiting `-P_x^⊥ ω / √(1-⟪x,ω⟫²)` as the
 ↪ Riemannian
gradient. -/
theorem inner_eq_inner_tangentialProjector {x ω x' : Eucl d}
    (htangent : (⟪x', x⟫ : ℝ) = 0) :
    (⟪x', ω⟫ : ℝ) = ⟪x', tangentialProjector x ω⟫ := by
  rw [tangentialProjector, inner_sub_right, real_inner_smul_right, htangent]
  ring
```

Leaves/GeodesicGradient.lean

2.4 Separating hyperplane

From the original paper (Geshkovski, Rigollet, Ruiz-Balet, Proposition 4.2 Step 1, p. 20). With ω_+ on the minimizing geodesic between ω and γ at distance $\pi/8$ from ω, one has $d_g(\omega_+, x_0^M) \geq d_g(\omega, x_0^M) - d_g(\omega, \omega_+) \geq \frac{3\pi}{8}$.
"As a consequence, the hyperplane $\{x \in \mathbb{S}^{d-1} : \langle \omega, x \rangle = \cos(\pi/8 + \tau)\}$ is a separating hyperplane for the ball $B(\omega, \pi/8 + \tau)$ and the points x_0^M and y^M for every $\tau \in (0, 3\pi/8)$; namely"

$$\langle \omega, x_0^M \rangle - \cos\Big(\frac{\pi}{8} + \tau\Big) = \cos d_g(\omega, x_0^M) - \cos\Big(\frac{\pi}{8} + \tau\Big) < 0,$$

"where the inequality is by virtue of (4.3)" (the choice $d_g(\omega, x_0^M) \geq \pi/2$).

Our formalization. machine-checked
The whole content is the strict antitonicity of cosine on $[0, \pi]$. From $\tau < 3\pi/8$ we get the cap angle $\pi/8 + \tau < \pi/2 \leq d_g(\omega, x)$, and both angles lie in $[0, \pi]$, so Mathlib's `Real.strictAntiOn_cos` gives $\cos d_g(\omega, x) < \cos(\pi/8 + \tau)$. Rewriting $\cos d_g(\omega, x) = \langle \omega, x \rangle$ lands the separating inequality $\langle \omega, x \rangle < \cos(\pi/8 + \tau)$. The key step is that monotonicity of cosine converts the angle gap $\pi/8 + \tau < d_g(\omega, x)$ into the inner-product gap that places x strictly on the far side of the hyperplane.

```
/-- L3 (separating side): if `x` is far from the anchor `ω` (`d_g(ω, x) ≥ π/2`) and
`τ ∈ (0, 3π/8)`, then `⟪ω, x⟫ < cos(π/8 + τ)`, so `x` lies strictly on the far side of
↪  the
separating hyperplane. -/
theorem separating_hyperplane {x ω : Eucl d} (hx : x ∈ sphere d) (hω : ω ∈ sphere d)
    {τ : ℝ} (hτ : τ ∈ Set.Ioo 0 (3 * Real.pi / 8))
    (hfar : Real.pi / 2 ≤ geodesicDist ω x) :
    ⟪ω, x⟫ < Real.cos (Real.pi / 8 + τ) := by
  obtain ⟨hτ0, hτ8⟩ := hτ
  have hpi := Real.pi_pos
  -- both angles lie in [0, π]
  have ha : Real.pi / 8 + τ ∈ Set.Icc (0 : ℝ) Real.pi := by
    constructor <;> nlinarith [Real.pi_pos]
  have hb : geodesicDist ω x ∈ Set.Icc (0 : ℝ) Real.pi := geodesicDist_mem_Icc ω x
  -- the far angle exceeds the cap angle
  have hab : Real.pi / 8 + τ < geodesicDist ω x := by nlinarith [hfar]
  -- cosine is strictly decreasing on [0, π]
  have hcos : Real.cos (geodesicDist ω x) < Real.cos (Real.pi / 8 + τ) :=
    Real.strictAntiOn_cos ha hb hab
  rwa [cos_geodesicDist hω hx] at hcos
```

`Leaves/SeparatingHyperplane.lean`

2.5 Lyapunov sign

From the original paper (Geshkovski, Rigollet, Ruiz-Balet, Example 6.1, pp. 29-30). On the circle ($d = 2$), for a particle following a characteristic of $\mu^i(t)$ one has $\dot{x}(t) = \mathbf{P}^{\perp}_{x(t)}(m_i(t)) = \alpha_i(t)\mathbf{P}^{\perp}_{x(t)}(u_i)$ and, in the angle variable $\vartheta := \theta - \theta_i$,

$$\dot{\vartheta}(t) = -\alpha_i(t) \sin\vartheta(t).$$

"The Lyapunov function $\mathsf{E}(\vartheta) := 1 - \cos\vartheta$ satisfies"

$$\dot{\mathsf{E}} = \sin\vartheta\,\dot{\vartheta} = -\alpha_i(t)\sin^2\vartheta \leq 0,$$

"with equality only at $\vartheta = 0$ (since $\alpha_i(t) > 0$ …). Thus $\vartheta(t) \to 0$ and each trajectory converges to u_i."

Our formalization. machine-checked
Both halves are kernel-checked. The derivative is the chain rule: $\frac{\mathrm{d}}{\mathrm{d}t}(1 - \cos\theta(s)) = \sin\theta(t)\,\theta'(t)$, and substituting the ODE $\theta' = -\alpha\sin\theta$ gives $-\alpha\sin^2\theta$. The sign lemma `lyapunov_deriv_nonpos` then notes $-\alpha\sin^2\theta \leq 0$ for $\alpha \geq 0$ because $\alpha\sin^2\theta \geq 0$. So $\mathsf{E} = 1 - \cos\theta$ is nonincreasing along the angle flow, forcing $\theta \to 0$. The key step is that the $\sin\theta$ produced by the ODE cancels with the $\sin\theta$ from differentiating $1 - \cos\theta$, leaving a perfect square of definite sign.

```
/-- L5 (derivative of the Lyapunov function along the flow): if `θ` solves `θ' = -α sin
 ↪  θ` at `t`,
then `E(s) = 1 - cos θ(s)` has derivative `-α sin²(θ t)` at `t`. -/
theorem lyapunov_hasDerivAt (α : ℝ) (θ : ℝ → ℝ) (t θ' : ℝ)
    (hθ : HasDerivAt θ θ' t) (hode : θ' = -α * Real.sin (θ t)) :
    HasDerivAt (fun s ⇒ 1 - Real.cos (θ s)) (-α * Real.sin (θ t) ^ 2) t := by
  have hcos : HasDerivAt (fun s ⇒ Real.cos (θ s)) (-Real.sin (θ t) * θ') t :=
    (Real.hasDerivAt_cos (θ t)).comp t hθ
  have h : HasDerivAt (fun s ⇒ 1 - Real.cos (θ s)) (0 - (-Real.sin (θ t) * θ')) t :=
    (hasDerivAt_const t 1).sub hcos
  convert h using 1
  rw [hode]; ring

/-- L5 (sign): the Lyapunov derivative is nonpositive when `α ≥ 0`. -/
theorem lyapunov_deriv_nonpos {α : ℝ} (hα : 0 ≤ α) (s : ℝ) :
    -α * Real.sin s ^ 2 ≤ 0 := by
  have : 0 ≤ α * Real.sin s ^ 2 := mul_nonneg hα (sq_nonneg _)
  linarith
```

Leaves/Lyapunov.lean

2.6 Ball-chain retention

From the original paper (Geshkovski, Rigollet, Ruiz-Balet, Lemma B.1 and its proof, pp. 31, 33).
Lemma B.1 transports mass through $K+1$ overlapping balls and concludes $\mu(T,\mathcal{B}_K) \geq (1-\varepsilon)^K \mu_0(\bigcup_k \mathcal{B}_k)$. Its proof:
"Write $[0,T] = \bigcup_{k\in[1,M]}[t_{k-1},t_k]$ where $t_k = kT/K$, and proceed by backward induction:"

$$\mu(T,\mathcal{B}_K) = \mu(T,\mathcal{B}_K \setminus \mathcal{B}_{K-1}) + \mu(T,\mathcal{B}_K \cap \mathcal{B}_{K-1}) \geq \mu(t_{K-1},\mathcal{B}_K \setminus \mathcal{B}_{K-1}) + (1-\varepsilon)\mu(t_{K-1},\mathcal{B}_{K-1}),$$

"where the last inequality follows from Lemma B.2. Using $\mathcal{B}_k \cap \mathcal{B}_{k'} = \emptyset$ whenever $|k-k'| \geq 2$, we arrive to"

$$\mu(T,\mathcal{B}_k) \geq (1-\varepsilon)^K \Big(\sum_{k=1}^{K} \mu_0(\mathcal{B}_k \setminus \mathcal{B}_{k-1}) + \mu_0(\mathcal{B}_0)\Big) = (1-\varepsilon)^K \mu_0\Big(\bigcup_{k\in[0,K]} \mathcal{B}_k\Big).$$

Our formalization. machine-checked
Stripped of the measure-theoretic bookkeeping, the backward induction is a clean arithmetic fact about a nonnegative sequence with per-step retention. We prove by induction on K that if $m \leq a_0$ and $(1-\varepsilon)\,a_k \leq a_{k+1}$ for all k, then $(1-\varepsilon)^K m \leq a_K$. The base case is $m \leq a_0$; the step chains $(1-\varepsilon)^{n+1} m = (1-\varepsilon)((1-\varepsilon)^n m) \leq (1-\varepsilon)\,a_n \leq a_{n+1}$. The key step is $0 \leq 1-\varepsilon$, which lets the induction hypothesis be scaled by $(1-\varepsilon)$ without reversing the inequality before applying the retention hypothesis.

```
/-- L9: geometric retention through a chain. With retention factor `1 - ε ∈ [0, 1]` per
 ↪ step and
initial mass at least `m ≥ 0`, after `K` steps at least `(1 - ε)^K · m` remains. -/
theorem ball_chain_geom {ε : ℝ} (hε1 : ε ≤ 1) (a : ℕ → ℝ) {m : ℝ}
    (h0 : m ≤ a 0) (hstep : ∀ k, (1 - ε) * a k ≤ a (k + 1)) :
    ∀ K : ℕ, (1 - ε) ^ K * m ≤ a K := by
  have h1ε : (0 : ℝ) ≤ 1 - ε := by linarith
  intro K
  induction K with
  | zero => simpa using h0
  | succ n ih =>
    calc (1 - ε) ^ (n + 1) * m
        = (1 - ε) * ((1 - ε) ^ n * m) := by ring
      _ ≤ (1 - ε) * a n := by exact mul_le_mul_of_nonneg_left ih h1ε
      _ ≤ a (n + 1) := hstep n
```

Leaves/BallChain.lean

2.7 Pigeonhole

From the original paper (Geshkovski, Rigollet, Ruiz-Balet, Lemma 3.4 Part 1, App. B.3, p. 35).
"Part 1. There exists an open ball $\mathcal{B} \subset \operatorname{supp} \mu_0 \cup \operatorname{supp} \nu_0$ such that $\mu_0(\mathcal{B}) \neq \nu_0(\mathcal{B})$. We now claim that there exists some $x^* \in \mathcal{B}$ such that"

$$\mu_0(\mathcal{B})x^* + \int_{\mathbb{S}^{d-1}\setminus\mathcal{B}} x\,\mu_0(\mathrm{d}x) \neq \nu_0(\mathcal{B})x^* + \int_{\mathbb{S}^{d-1}\setminus\mathcal{B}} x\,\nu_0(\mathrm{d}x).$$

"Indeed if this were to be false, then we'd have"

$$x^* = \frac{1}{\mu_0(\mathcal{B}) - \nu_0(\mathcal{B})} \int_{\mathbb{S}^{d-1}\setminus\mathcal{B}} x\,(\nu_0(\mathrm{d}x) - \mu_0(\mathrm{d}x))$$

"for all $x^* \in \mathcal{B}$, which cannot hold."

Our formalization. machine-checked
The displayed contradiction is "x^* equals a fixed vector for all $x^* \in \mathcal{B}$" — i.e. the identity would be constant on the open ball $\mathcal{B}$, impossible. We isolate exactly that general fact: in a nontrivial real normed space, an open ball of positive radius contains a point distinct from any prescribed a. Pick $u \neq 0$ and set $v = (r/2\|u\|)\,u$, so $\|v\| = r/2 < r$ and $v \neq 0$; then c and $c+v$ both lie in $B(c,r)$ (at distances 0 and $r/2$) and are distinct. If $c = a$, return $c+v$ (which differs from $c = a$ because $v \neq 0$); otherwise return c. The key step is that a positive-radius ball is never a single point, so it cannot be forced to one value — the contradiction the pigeonhole argument exploits.

```lean
/-- L10: an open ball of positive radius in a nontrivial real normed space contains a
 ↪  point distinct
from any prescribed `a`. Hence no map can be forced to a single value on the ball, which
 ↪  is the
contradiction the pigeonhole step exploits. -/
theorem exists_ne_in_ball (c a : E) {r : ℝ} (hr : 0 < r) :
    ∃ x ∈ Metric.ball c r, x ≠ a := by
  obtain ⟨u, hu⟩ := exists_ne (0 : E)
  have hunorm : 0 < ‖u‖ := norm_pos_iff.mpr hu
  set v : E := (r / (2 * ‖u‖)) • u with hv
  have hvnorm : ‖v‖ = r / 2 := by
    rw [hv, norm_smul, Real.norm_of_nonneg (by positivity)]
    field_simp
  have hvne : v ≠ 0 := by
    rw [← norm_pos_iff, hvnorm]; linarith
  -- `c` and `c + v` both lie in the ball and are distinct
  have hc_mem : c ∈ Metric.ball c r := Metric.mem_ball_self hr
  have hcv_mem : c + v ∈ Metric.ball c r := by
    rw [Metric.mem_ball, dist_eq_norm]
    have h1 : c + v - c = v := by abel
    rw [h1, hvnorm]; linarith
  -- at least one of the two distinct points differs from `a`
  rcases eq_or_ne c a with hca | hca
  · refine ⟨c + v, hcv_mem, fun h => hvne ?_⟩
    have h2 : c + v = c + 0 := by rw [add_zero, h, hca]
    exact add_left_cancel h2
  · exact ⟨c, hc_mem, hca⟩
```

Leaves/Pigeonhole.lean

2.8 Disjoint hulls give non-colinear barycenters

From the original paper (Geshkovski, Rigollet, Ruiz-Balet, Proposition 3.1 induction step, pp. 16-17).
The induction hypothesis (3.3) is

$$\operatorname{conv}_g \operatorname{supp} \mu_0^i \cap \operatorname{conv}_g \operatorname{supp} \mu_0^j = \emptyset \qquad \text{for } i \neq j \in [\![1, N-1]\!]. \tag{3.3}$$

The induction step then asserts, without proof:
"Since $\operatorname{supp} \mu_0^i \subset \mathbb{Q}_1^{d-1}$, (3.3) implies that"

$$\mathbb{E}_{\mu_0^i}[x] \text{ is not colinear with } \mathbb{E}_{\mu_0^j}[x] \qquad \text{for } i \neq j \in [\![1, N-1]\!].$$

Our formalization. machine-checked
Within an open hemisphere spherical geodesics are radial projections of Euclidean chords, so we model the geodesic hull as $\operatorname{conv}_g(s) = \operatorname{cone}(s) \cap \mathbb{S}^{d-1}$ — the unit vectors that are nonnegative combinations of s (`geodesicHull`). The empirical barycenter $\sum_p w_p\, p$ with $w_p \geq 0$ is a

manifest conical combination of s, and the cone is closed under nonnegative scaling, so the normalization $\|b\|^{-1}b$ of any nonzero conical point lies in $\mathrm{conv}_g(s)$. If the two barycenters were colinear — for vectors in the positive orthant "colinear" means a *positive* multiple, i.e. `SameRay` — then $b = k\,c$ with $k > 0$, so their normalizations coincide, $\|b\|^{-1}b = \|c\|^{-1}c$. That common unit direction would then sit in *both* geodesic hulls, contradicting their disjointness. The key step is that positive colinearity forces a shared normalized direction, which disjoint hulls forbid.

```
/-- The geodesic-convex hull of `s`, modeled within a hemisphere as `cone(s) ∩ 𝕊^{d-1}`:
↪ the unit
vectors that are nonnegative combinations of `s`. Faithful to `conv_g` when `s` lies in
↪ an open
hemisphere (e.g. the orthant `Q₁^{d-1}`), since spherical geodesics are radial
↪ projections of chords. -/
def geodesicHull (s : Finset (Eucl d)) : Set (Eucl d) :=
  {x | ‖x‖ = 1 ∧ inConicalSpan s x}

/-- **Core of finding F2.** If two nonzero conical points `b`, `c` (the unnormalized
↪ barycenters) are
colinear — `SameRay ℝ b c`, i.e. positive multiples, which is what "colinear" means for
↪ vectors in the
positive orthant — then their common normalized direction lies in *both* geodesic hulls.
↪ -/
theorem geodesicHull_inter_nonempty_of_sameRay {s₁ s₂ : Finset (Eucl d)} {b c : Eucl d}
    (hb : inConicalSpan s₁ b) (hc : inConicalSpan s₂ c) (hb0 : b ≠ 0) (hc0 : c ≠ 0)
    (hsr : SameRay ℝ b c) : (geodesicHull s₁ ∩ geodesicHull s₂).Nonempty := by
  obtain ⟨r₁, r₂, hr₁, hr₂, hrel⟩ := hsr.exists_pos hb0 hc0
  -- `b = k • c` with `k = r₁⁻¹ r₂ > 0`
  set k : ℝ := r₁⁻¹ * r₂ with hkdef
  have hk : 0 < k := by positivity
  have hbk : b = k • c := by
    rw [hkdef, mul_smul, ← hrel, smul_smul, inv_mul_cancel₀ (ne_of_gt hr₁), one_smul]
  -- the normalized vectors coincide
  have key : ‖b‖⁻¹ • b = ‖c‖⁻¹ • c := by
    have hnc : ‖c‖ ≠ 0 := norm_ne_zero_iff.mpr hc0
    rw [hbk, norm_smul, Real.norm_eq_abs, abs_of_pos hk, smul_smul]
    congr 1
    field_simp
  refine ⟨‖b‖⁻¹ • b, ?_⟩
  rw [Set.mem_inter_iff]
  exact ⟨mem_geodesicHull_normalize hb hb0, key ▸ mem_geodesicHull_normalize hc hc0⟩

/-- **Finding F2, discharged.** Empirical measures (with nonnegative weights) in the
↪ orthant whose
geodesic hulls are disjoint have non-colinear barycenters. This is the step Proposition
↪ 3.1's
induction asserts without proof. -/
theorem barycenter_noncolinear_of_disjoint_hull {s₁ s₂ : Finset (Eucl d)} {w₁ w₂ : Eucl
↪ d → ℝ}
    (hw₁ : ∀ p ∈ s₁, 0 ≤ w₁ p) (hw₂ : ∀ p ∈ s₂, 0 ≤ w₂ p)
    (hb0 : (∑ p ∈ s₁, w₁ p • p) ≠ 0) (hc0 : (∑ p ∈ s₂, w₂ p • p) ≠ 0)
    (hdisj : Disjoint (geodesicHull s₁) (geodesicHull s₂)) :
    ¬ SameRay ℝ (∑ p ∈ s₁, w₁ p • p) (∑ p ∈ s₂, w₂ p • p) := by
  intro hsr
  obtain ⟨u, hu⟩ := geodesicHull_inter_nonempty_of_sameRay
    (barycenter_mem_conicalSpan hw₁) (barycenter_mem_conicalSpan hw₂) hb0 hc0 hsr
  rw [Set.mem_inter_iff] at hu
  exact Set.disjoint_left.mp hdisj hu.1 hu.2
```

Leaves/BarycenterNonColinear.lean

Strengthening (finding F2). Proposition 3.1's induction *asserts* "(3.3) [disjoint geodesic hulls] $\Rightarrow \mathbb{E}_{\mu_0^i}[x]$ not colinear with $\mathbb{E}_{\mu_0^j}[x]$" with no justification — the one genuine rigor gap the adversarial review flagged. This leaf supplies the kernel-checked proof for the empirical regime (the Dirac targets of Theorem 1.1 and the empirical targets of the restricted Theorem 1.2), under the faithful hemisphere model $\mathrm{conv}_g(s) = \mathrm{cone}(s) \cap \mathbb{S}^{d-1}$.

2.9 F2, general probability measures

From the original paper (Geshkovski, Rigollet, Ruiz-Balet, Proposition 3.1 induction step, pp. 16-17).
The same unproved assertion as above, now read for arbitrary $\mu_0, \nu_0 \in \mathcal{P}(\mathbb{Q}_1^{d-1})$ with barycenter $\mathbb{E}_{\mu_0}[x] = \int x \, \mu_0(\mathrm{d}x)$:

$$\mathrm{conv}_g \operatorname{supp} \mu_0 \cap \mathrm{conv}_g \operatorname{supp} \nu_0 = \emptyset \implies \mathbb{E}_{\mu_0}[x] \text{ not colinear with } \mathbb{E}_{\nu_0}[x].$$

Our formalization. machine-checked
The finite case above is elementary; for a general probability measure the only extra ingredient is the standard fact that the barycenter $\int x \, \mathrm{d}\mu$ lies in any closed convex set carrying full mass — Mathlib's `Convex.integral_mem`. Wrapping it as `barycenter_mem_of_supportedIn` (from $\mu(s^c) = 0$ to $\int x \, \mathrm{d}\mu \in s$), the same `SameRay` argument runs verbatim: both normalized barycenters land in their respective closed convex *cones* s_1, s_2, the shared unit direction sits in both, and disjointness on the sphere is contradicted. The key step is that `Convex.integral_mem` is the measure-theoretic analogue of "the barycenter lies in the convex hull of the support," so no optimal-transport axiom is reintroduced.

```
/-- The barycenter of a probability measure lies in any closed convex set carrying full
↪ mass
(`μ Sᶜ = 0`). This is the standard measure-theoretic fact the general case of finding F2
↪ needs; it
rests only on Mathlib's `Convex.integral_mem`, not on the optimal-transport axioms. -/
theorem barycenter_mem_of_supportedIn {μ : Measure (Eucl d)} [IsProbabilityMeasure μ]
    (hint : Integrable (fun x : Eucl d => x) μ) {s : Set (Eucl d)}
    (hconv : Convex ℝ s) (hclosed : IsClosed s) (hsupp : μ sᶜ = 0) :
    barycenter μ ∈ s := by
  have hae : ∀ᵐ x ∂μ, x ∈ s := by
    rw [ae_iff]; exact hsupp
  rw [barycenter]
  exact hconv.integral_mem hclosed hae hint

/-- **Finding F2, general case.** Two probability measures whose supports lie in
↪ disjoint closed
convex cones `s₁`, `s₂` (disjoint on the sphere) have non-colinear barycenters.
↪ Generalizes
`barycenter_noncolinear_of_disjoint_hull` from empirical to arbitrary probability
↪ measures; kernel-
clean, via `barycenter_mem_of_supportedIn`. -/
theorem barycenter_noncolinear_of_disjoint_hull_general
    {μ ν : Measure (Eucl d)} [IsProbabilityMeasure μ] [IsProbabilityMeasure ν]
    (hμi : Integrable (fun x : Eucl d => x) μ) (hνi : Integrable (fun x : Eucl d => x) ν)
    {s₁ s₂ : Set (Eucl d)}
    (hc₁ : Convex ℝ s₁) (hcl₁ : IsClosed s₁) (hcone₁ : ∀ x ∈ s₁, ∀ a : ℝ, 0 ≤ a → a • x ∈
↪   s₁)
    (hc₂ : Convex ℝ s₂) (hcl₂ : IsClosed s₂) (hcone₂ : ∀ x ∈ s₂, ∀ a : ℝ, 0 ≤ a → a • x ∈
↪   s₂)
    (hμs : μ s₁ᶜ = 0) (hνs : ν s₂ᶜ = 0)
    (hb0 : barycenter μ ≠ 0) (hc0 : barycenter ν ≠ 0)
    (hdisj : ∀ u : Eucl d, ‖u‖ = 1 → u ∈ s₁ → u ∉ s₂) :
    ¬ SameRay ℝ (barycenter μ) (barycenter ν) := by
  intro hsr
  set b := barycenter μ with hbdef
  set c := barycenter ν with hcdef
  obtain ⟨r₁, r₂, hr₁, hr₂, hrel⟩ := hsr.exists_pos hb0 hc0
  have hbs₁ : b ∈ s₁ := barycenter_mem_of_supportedIn hμi hc₁ hcl₁ hμs
  have hcs₂ : c ∈ s₂ := barycenter_mem_of_supportedIn hνi hc₂ hcl₂ hνs
  set k : ℝ := r₁⁻¹ * r₂ with hkdef
  have hk : 0 < k := by positivity
  have hbk : b = k • c := by
    rw [hkdef, mul_smul, ← hrel, smul_smul, inv_mul_cancel₀ (ne_of_gt hr₁), one_smul]
  have key : ‖b‖⁻¹ • b = ‖c‖⁻¹ • c := by
    have hnc : ‖c‖ ≠ 0 := norm_ne_zero_iff.mpr hc0
    rw [hbk, norm_smul, Real.norm_eq_abs, abs_of_pos hk, smul_smul]
    congr 1
    field_simp
  set u := ‖b‖⁻¹ • b with hu
  have hunorm : ‖u‖ = 1 := by
    rw [hu, norm_smul, norm_inv, norm_norm, inv_mul_cancel₀ (norm_ne_zero_iff.mpr hb0)]
  have hus₁ : u ∈ s₁ := by rw [hu]; exact hcone₁ b hbs₁ _ (by positivity)
  have hus₂ : u ∈ s₂ := by rw [key]; exact hcone₂ c hcs₂ _ (by positivity)
  exact hdisj u hunorm hus₁ hus₂
```

Leaves/BarycenterNonColinear.lean

Strengthening (finding F2, residual). The finite leaf above closes F2 for the empirical regime; this companion closes the remaining general-measure case, extending the non-colinearity to arbitrary probability measures supported in disjoint closed convex cones. It rests only on Mathlib's `Convex.integral_mem` and introduces **no new axioms** — in particular it does not reach for the optimal-transport (W_2) axiom.

2.10 Linearized OT bound

From the original paper (Geshkovski, Rigollet, Ruiz-Balet, Lemma 5.2, p. 24).
"Lemma 5.2. Suppose $\mu \in \mathcal{P}(\mathbb{S}^{d-1})$ and $T^1, T^2 : \mathbb{S}^{d-1} \to \mathbb{S}^{d-1}$ measurable, with T^1 bijective. Then"

$$\mathsf{W}_2(T^1_\# \mu,\, T^2_\# \mu) \lesssim \|T^1 - T^2\|_{L^2(\mu)}. \tag{5.2}$$

"Proof of Lemma 5.2. Since T^1 is bijective, there is a measurable $\psi : \mathbb{S}^{d-1} \to \mathbb{S}^{d-1}$ such that $\psi(T^1(x)) = T^2(x)$ for all $x \in \mathbb{S}^{d-1}$. Then"

$$\mathsf{W}_2^2(T^1_\# \mu,\, T^2_\# \mu) \lesssim \int \|x - \psi(x)\|^2 (T^1_\# \mu)(\mathrm{d}x) = \|T^1 - T^2\|^2_{L^2(\mu)}.$$

Our formalization. axiomatised
This is the map-induced coupling bound. Since T^1 is bijective there is ψ with $\psi \circ T^1 = T^2$, and the deterministic coupling $(\mathrm{id}, \psi)_\# (T^1_\# \mu)$ between the two pushforwards realizes transport cost $\int \|x - \psi(x)\|^2 \,\mathrm{d}(T^1_\# \mu) = \|T^1 - T^2\|^2_{L^2(\mu)}$, bounding W_2^2. Mathlib `v4.31.0` has no W_2, so the Lean statement is asserted over the labeled optimal-transport axiom `W2_map_le_L2`; the node is therefore **axiomatised**, not a kernel proof. Its effective status is the minimum over its dependency closure, which contains the W_2 axiom.

```
/-- L7 (Lemma 5.2): the `W₂` distance between two pushforwards of `μ` is bounded by the
 ↪  `L²(μ)`
distance of the maps. Rests on the optimal-transport axiom `W2_map_le_L2`. -/
theorem lemma_5_2 (μ : Measure (Eucl d)) (T₁ T₂ : Eucl d → Eucl d) :
    W2 (μ.map T₁) (μ.map T₂) ≤ Real.sqrt (∫ x, ‖T₁ x - T₂ x‖ ^ 2 ∂μ) :=
  W2_map_le_L2 μ T₁ T₂
```

Leaves/Coupling.lean

2.11 Markov bound

From the original paper (Geshkovski, Rigollet, Ruiz-Balet, Claim 2, App. B.5, p. 39).
"Claim 2. Suppose $\mu \in \mathcal{P}(\mathbb{S}^{d-1})$ and $x_0 \in \mathbb{S}^{d-1}$ satisfy $\mathsf{W}_2(\mu, \delta_{x_0}) \leq \eta_2$. There exists a universal constant $C > 0$ such that $1 - \mu(B(x_0, \eta_3)) \leq C\eta_2/\eta_3$ for all $\eta_3 > 0$."
"Proof of Claim 2. By compactness of $\mathbb{S}^{d-1}$ and Kantorovich-Rubinstein duality,"

$$\mathsf{W}_1(\mu, \delta_{x_0}) = \sup_{\mathrm{Lip}(g)\leq 1} \int g\,(\mu - \delta_{x_0}) \leq C \cdot \eta_2$$

"for some numerical constant $C > 0$. Hence, for $g : \mathbb{S}^{d-1} \to \mathbb{S}^{d-1}$ defined as"

$$g(x) = \begin{cases} 1 & x \in B(x, \eta_3) \\ 1 - \dfrac{1-\eta_3}{\eta_3} & x \in B(x, \eta_3) \cap B(x, 2\eta_3) \\ 0 & x \notin B(x, 2\eta_3), \end{cases}$$

"we obtain $1 - \mu(B(x_0, \eta_3)) \leq C\eta_2/\eta_3$."

Our formalization. axiomatised
We take the cleaner truncated-distance bump $f(x) = \min(\eta_3,\ d(x, x_0))$ in place of the paper's hat. It is **machine-checked 1-Lipschitz** — `distBump_lipschitz`, obtained by truncating the 1-Lipschitz distance through the map $t \mapsto \min(\eta_3, t)$ and routed through the `dist`-characterization to dodge the `EuclideanSpace/PiLp` extended-metric instance diamond — vanishes at x_0, and dominates $\eta_3 \cdot \mathbf{1}_{\{d(\cdot,x_0)\geq\eta_3\}}$. Hence $\eta_3\,\mu\{d(\cdot, x_0) \geq \eta_3\} \leq \int f\,\mathrm{d}\mu = \int f\,\mathrm{d}\mu - f(x_0) \leq \mathsf{W}_1(\mu, \delta_{x_0}) \leq C\eta_2$, and since $\{\eta_3 \leq d(\cdot, x_0)\} = B(x_0, \eta_3)^c$ this is $1 - \mu(B(x_0, \eta_3)) \leq C\eta_2/\eta_3$ for a probability measure. The Lipschitz bound is kernel-checked, but the middle inequality is Kantorovich-Rubinstein duality, supplied here by the labeled W_1 axiom `W1_ge_of_lipschitz`; so the node as a whole is **axiomatised**.

```
/-- The truncated-distance bump `f(x) = min(η₃, d(x, x₀))`. -/
noncomputable def distBump (x₀ : Eucl d) (η₃ : ℝ) (x : Eucl d) : ℝ := min η₃ (dist x x₀)

/-- The bump is `1`-Lipschitz (a truncation of the `1`-Lipschitz distance). Proved
 ↪  through the
`dist`-characterization to avoid the `EuclideanSpace`/`PiLp` extended-metric instance
 ↪  diamond. -/
theorem distBump_lipschitz (x₀ : Eucl d) (η₃ : ℝ) : LipschitzWith 1 (distBump x₀ η₃) :=
 ↪  by
  rw [lipschitzWith_iff_dist_le_mul]
  intro x y
  -- `t ↦ min η₃ t` is 1-Lipschitz on ℝ (no PiLp instance diamond on the codomain)
  have hL : LipschitzWith 1 (fun t : ℝ => min η₃ t) := LipschitzWith.const_min
 ↪  LipschitzWith.id η₃
  have hmin := hL.dist_le_mul (dist x x₀) (dist y x₀)
  simp only [distBump, Real.dist_eq, NNReal.coe_one, one_mul] at hmin ⊢
  exact hmin.trans (abs_dist_sub_le x y x₀)

/-- **Lemma L8 / Claim 2 (Markov bound).** For a probability measure with
`W₁(μ, δ_{x₀}) ≤ C η₂`, the mass at geodesic/Euclidean distance `≥ η₃` from `x₀` is at
 ↪  most
`C η₂ / η₃`. Equivalently `1 - μ(B(x₀, η₃)) ≤ C η₂ / η₃`. Rests on the `W₁` axiom (KR
 ↪  duality):
status `math.axiomatised`. -/
theorem markov_bound (μ : Measure (Eucl d)) [IsProbabilityMeasure μ]
    (x₀ : Eucl d) (η₂ η₃ C : ℝ) (hη₃ : 0 < η₃)
    (hW1 : W1 μ (Measure.dirac x₀) ≤ C * η₂) :
    μ.real {x | η₃ ≤ dist x x₀} ≤ C * η₂ / η₃ := by
  set s : Set (Eucl d) := {x | η₃ ≤ dist x x₀} with hs
  have hcont : Continuous (fun x : Eucl d => dist x x₀) := by fun_prop
  have hsmeas : MeasurableSet s := measurableSet_le measurable_const hcont.measurable
  -- the bump is integrable (bounded by η₃ on a finite measure)
  have hbump_int : Integrable (distBump x₀ η₃) μ := by
    refine Integrable.mono' (integrable_const η₃)
      (distBump_lipschitz x₀ η₃).continuous.aestronglyMeasurable ?_
    filter_upwards with x
    rw [Real.norm_eq_abs, abs_of_nonneg (distBump_nonneg hη₃.le x)]
    exact min_le_left _ _
  have hind_int : Integrable (s.indicator (fun _ => η₃)) μ :=
    (integrable_const η₃).indicator hsmeas
  -- η₃ · 𝟙_s ≤ bump pointwise
  have hdom : s.indicator (fun _ => η₃) ≤ distBump x₀ η₃ := by
    intro x
    by_cases hx : x ∈ s
    · rw [Set.indicator_apply, if_pos hx]; exact (distBump_eq_outside hx).ge
    · rw [Set.indicator_apply, if_neg hx]; exact distBump_nonneg hη₃.le x
  -- ∫ η₃·𝟙_s dμ = η₃ · μ(s)
  have hint_ind : ∫ x, s.indicator (fun _ => η₃) x ∂μ = μ.real s * η₃ := by
    rw [integral_indicator_const η₃ hsmeas, smul_eq_mul]
  -- ∫ bump dδ_{x₀} = f(x₀) = 0
  have hdirac : ∫ x, distBump x₀ η₃ x ∂(Measure.dirac x₀) = 0 := by
    rw [integral_dirac]
    simp [distBump, min_eq_right hη₃.le]
  -- KR duality (axiom): ∫ bump dμ - ∫ bump dδ ≤ W₁
  have hdual := W1_ge_of_lipschitz μ (Measure.dirac x₀) (distBump x₀ η₃)
 ↪  (distBump_lipschitz x₀ η₃)
  rw [hdirac, sub_zero] at hdual
  -- assemble: η₃ · μ(s) ≤ ∫ bump dμ ≤ W₁ ≤ C η₂
  have hmono : μ.real s * η₃ ≤ ∫ x, distBump x₀ η₃ x ∂μ := by
```

```
Leaves/MarkovBound.lean
```

3 Mass transport and clustering

The ReLU-gated drift $\mathbf{W}(\mathbf{U}x+b)_+$ transports mass through a chain of overlapping balls, retaining a $(1-\varepsilon)^K$ fraction of it after K parameter switches. Self-attention then clusters a measure supported in a single hemisphere to a point, and chaining the two steers an ensemble of disjoint-support atomless measures to prescribed discrete targets.

3.1 Single overlapping ball pair (Lemma B.2)

From the original paper (Geshkovski, Rigollet, Ruiz-Balet, Lemma B.2, p. 31). Consider two open balls $\mathcal{B}_0, \mathcal{B}_1 \subset \mathbb{S}^{d-1}$ such that $\mathcal{B}_0 \cap \mathcal{B}_1 \neq \emptyset$. For any $\varepsilon > 0$ and $T > 0$, there exist $\mathbf{W}, \mathbf{U} \in \mathcal{M}_{d\times d}(\mathbb{R})$ and $b \in \mathbb{R}^d$ such that for any $\mu_0 \in \mathcal{P}(\mathbb{S}^{d-1})$, the unique solution μ to (B.1) satisfies

$$\mu(T, \mathcal{B}_0 \cap \mathcal{B}_1) \geq (1-\varepsilon)\,\mu_0(\mathcal{B}_0). \tag{B.3}$$

Moreover $\mu(T) = \phi^T_{\#}\mu_0$ where the Lipschitz-continuous and invertible flow map $\Phi^t : \mathbb{S}^{d-1} \to \mathbb{S}^{d-1}$ of (4.1) satisfies, for all $t \in [0, T]$,

$$(\phi^t)_{|\mathbb{S}^{d-1}\setminus\mathcal{B}_0} \equiv \mathrm{Id}\,.$$

Furthermore, for any fixed $\omega \in \operatorname{int}\mathcal{B}_0$ we can choose $\mathbf{W}, \mathbf{U}$ and b so that the solution to (B.1) satisfies $W_2(\mu(T), \alpha) \leq \varepsilon$ where $\alpha(A) = \mu_0(\mathcal{B}_0)\,\delta_\omega(A) + \mu_0(A \setminus \mathcal{B}_0)$, for any Borel $A \subset \mathbb{S}^{d-1}$. *Proof sketch (full proof p. 32).* Let z and R be the center and radius of $\mathcal{B}_0$ and fix $\omega \in \operatorname{int}(\mathcal{B}_0 \cap \mathcal{B}_1)$. The authors choose $\mathbf{U}, b$ and a $\mathbf{W}$ with $\mathbf{W}\mathbf{1} = \omega$ so that the gated drift is $(-\cos d_g(z,x) + \cos R)_+\,\omega = (\cos R - \langle z, x\rangle)_+\,\omega$, and record (B.4) $(\cos R - \langle z, x\rangle)_+ > 0 \iff x \in \mathcal{B}_0$ (see the strengthening below). Then (B.5)

$$\tfrac{d}{dt}\langle x(t), \omega\rangle = (\cos R - \langle z, x(t)\rangle)_+(1 - \langle x(t), \omega\rangle^2),$$

which is positive whenever $x(t) \in \mathcal{B}_0 \setminus \{\omega\}$. Hence for small $\delta > 0$ there is a time T_δ with $\phi^{T_\delta}(x) \in \mathcal{B}_0 \cap \mathcal{B}_1$ for $x \in B(z, R-\delta)$ (B.7); pushing forward, $\mu(T_\delta, \mathcal{B}_0 \cap \mathcal{B}_1) = \mu_0((\phi^{T_\delta})^{-1}(\mathcal{B}_0 \cap \mathcal{B}_1)) \geq \mu_0(B(z, R-\delta))$ (B.8), and taking δ small enough that $\mu_0(B(z, R-\delta)) \geq (1-\varepsilon)\mu_0(\mathcal{B}_0)$ gives (B.3) after rescaling time. The gate vanishes off $\mathcal{B}_0$, so $\phi^t \equiv \mathrm{Id}$ there. The bound to α follows by taking $\mathcal{B}_1 = B(\omega, \eta) \subset \mathcal{B}_0$ and a W_1 Kantorovich-Rubinstein estimate, sending η and the slack to zero.

What we strengthen. As printed the proof takes $\mathbf{U} = -\mathbf{1}z^\top$, $b = \cos(R)\mathbf{1}$ (with $\mathbf{W}\mathbf{1} = \omega$), so $\mathbf{U}x = -\langle z, x\rangle\mathbf{1}$, giving the gate coefficient $(\cos R - \langle z, x\rangle)_+$, and asserts in (B.4) that this is positive iff $x \in \mathcal{B}_0 = B(z, R)$. But

$$\cos R - \langle z, x\rangle > 0 \iff \langle z, x\rangle < \cos R \iff d_g(z, x) > R \iff x \notin \mathcal{B}_0,$$

so as written the gate is active on the **complement** of $\mathcal{B}_0$: the interior mass it is meant to move sees a zero gate, and the lemma transports nothing. The fix is a one-sign typo — take $\mathbf{U} = +\mathbf{1}z^\top$, $b = -\cos(R)\mathbf{1}$, giving the gate $(\langle z, x\rangle - \cos R)_+$, positive exactly on $\mathcal{B}_0$ — after which (B.3)-(B.5) and the rest of the argument go through and Lemma B.2 holds as stated. The corrected equivalence $g(x) > 0 \iff d_g(z, x) > R$ is the kernel-checked leaf **L2** (`gate_pos_iff_dist`, machine-checked); the same inversion is corroborated independently by Proposition 4.2, Step 3 (which states the identical construction with the correct activation side) and by experiment E1. This is finding **F1**; see ERRATA.md.

Our formalization. open

The authors' strategy is the gated-drift construction above: a single ReLU gate, active on $\mathcal{B}_0$ (after the F1 sign correction), pushes the interior mass of $\mathcal{B}_0$ along $+\omega$ into $\mathcal{B}_0 \cap \mathcal{B}_1$ while leaving everything off $\mathcal{B}_0$ fixed, so a $(1-\varepsilon)$ fraction is retained in the overlap. We state this type-correctly in Lean against the labeled axiom layer — the solution map `measureFlow`, the switch count `switches`, and the Wasserstein distance `W2` — with "no mass outside S" read measure-theoretically. It is an open `sorry` stub: we do **not** claim a proof; discharging it rests on the continuity-equation flow axiom.

```
/-- **Lemma B.2** (single ball pair). Mass in `B₀` is pushed into `B₀ ∩ B₁`, retaining
 ↪ `(1-ε)`. Stub
(`sorry`): `math.open`. (Review finding F1: the paper's printed gate parameters have
 ↪ the activation
side reversed; the corrected sign is `U = +𝟙 zᵀ, b = -cos(R) 𝟙`. See `ERRATA.md`.) -/
theorem lemma_B_2 (μ : Measure (Eucl d)) (T ε : ℝ) (hT : 0 < T) (hε : 0 < ε)
    (B₀ B₁ : Set (Eucl d)) (hcap : (B₀ ∩ B₁).Nonempty) :
    ∃ θ : Params d, (1 - ENNReal.ofReal ε) * μ B₀ ≤ (measureFlow θ T μ) (B₀ ∩ B₁) := by
  sorry
```

Statements/MidLevel.lean

3.2 Chain of overlapping balls (Lemma B.1)

From the original paper (Geshkovski, Rigollet, Ruiz-Balet, Lemma B.1, p. 31).
Consider $K+1$ open balls $\mathcal{B}_K, \dots, \mathcal{B}_1, \mathcal{B}_0 \subset \mathbb{S}^{d-1}$ satisfying

$$\mathcal{B}_k \cap \mathcal{B}_{k-1} \neq \emptyset \quad \text{for } k \in [\![1, K]\!], \qquad \mathcal{B}_k \cap \mathcal{B}_{k'} = \emptyset \quad \text{if } |k - k'| \geq 2.$$

Then for any $T > 0$ and $\varepsilon > 0$, there exist $(\mathbf{W}, \mathbf{U}, b) : [0, T] \to \mathcal{M}_{d\times d}(\mathbb{R})^2 \times \mathbb{R}^d$, piecewise constant having at most K switches, such that for any $\mu_0 \in \mathcal{P}(\mathbb{S}^{d-1})$, the corresponding unique solution μ to

$$\begin{cases} \partial_t \mu(t) + \operatorname{div}(\mathbf{P}_x^\perp \mathbf{W}(t)(\mathbf{U}(t)x + b(t))_+ \mu(t)) = 0 & \text{on } [0,T] \times \mathbb{S}^{d-1}, \\ \mu(0) = \mu_0 & \text{on } \mathbb{S}^{d-1}, \end{cases} \tag{B.1}$$

satisfies

$$\mu(T, \mathcal{B}_K) \geq (1-\varepsilon)^K \mu_0\Big(\bigcup_k \mathcal{B}_k\Big).$$

Moreover, $\mu(T) = \phi^T_\# \mu_0$ for a Lipschitz-continuous, invertible map $\phi^t : \mathbb{S}^{d-1} \to \mathbb{S}^{d-1}$ which satisfies, for all $t \in [0, T]$,

$$\phi^t(x) = x \quad \text{for } x \notin \bigcup_k \mathcal{B}_k. \tag{B.2}$$

Proof sketch (full proof p. 33). Backward induction on the chain. Write $[0, T) = \bigcup_{k \in [\![1,K]\!]} [t_{k-1}, t_k)$ with $t_k = kT/K$ and apply Lemma B.2 to the adjacent pair $(\mathcal{B}_{k-1}, \mathcal{B}_k)$ on each subinterval. Then

$$\mu(T, \mathcal{B}_K) = \mu(T, \mathcal{B}_K \setminus \mathcal{B}_{K-1}) + \mu(T, \mathcal{B}_K \cap \mathcal{B}_{K-1}) \geq \mu(t_{K-1}, \mathcal{B}_K \setminus \mathcal{B}_{K-1}) + (1-\varepsilon)\, \mu(t_{K-1}, \mathcal{B}_{K-1}),$$

the last step by Lemma B.2. The non-adjacency disjointness $\mathcal{B}_k \cap \mathcal{B}_{k'} = \emptyset$ for $|k - k'| \geq 2$ keeps the steps from interfering, and iterating gives

$$\mu(T, \mathcal{B}_K) \geq (1-\varepsilon)^K \Big(\sum_{k=1}^{K} \mu_0(\mathcal{B}_k \setminus \mathcal{B}_{k-1}) + \mu_0(\mathcal{B}_0) \Big) = (1-\varepsilon)^K \mu_0 \Big(\bigcup_{k \in [\![0,K]\!]} \mathcal{B}_k \Big).$$

Our formalization. open

The authors' argument is a backward induction that applies the single-pair lemma (Lemma B.2) once per adjacent overlap, each step retaining a factor $(1-\varepsilon)$, the non-adjacency disjointness guaranteeing the steps do not interfere. We state the retention bound type-correctly in Lean against the axiom layer (`measureFlow`, `switches`); it is an open `sorry` stub. The purely arithmetic heart — that a per-step retention factor $(1-\varepsilon)$ compounds to $a_K \geq (1-\varepsilon)^K a_0$ — is already discharged kernel-clean by leaf **L9** (`ball_chain_geom`, machine-checked); what remains open is the geometric non-interference, which rests on the parking property of the flow axiom.

```
/-- **Lemma B.1** (ball-chain retention). For a chain of `K+1` overlapping balls, `K`
↪ switches retain
`(1-ε)^K` of the mass into the last ball. Stub (`sorry`): `math.open`. The arithmetic
↪ core is leaf
L9; the geometric non-interference is the parking axiom. -/
theorem lemma_B_1 (μ : Measure (Eucl d)) (T ε : ℝ) (hT : 0 < T) (hε : 0 < ε)
    (K : ℕ) (B : ℕ → Set (Eucl d)) :
    ∃ θ : Params d, switches θ ≤ K ∧
      (1 - ENNReal.ofReal ε) ^ K * μ (⋃ k, B k) ≤ (measureFlow θ T μ) (B K) := by
  sorry
```

Statements/MidLevel.lean

3.3 Clustering a single-hemisphere measure to a point (Proposition 2.1)

From the original paper (Geshkovski, Rigollet, Ruiz-Balet, Proposition 2.1, p. 11). Suppose $\mathbf{B} \in \mathcal{M}_{d\times d}(\mathbb{R})$ and $\operatorname{supp} \mu_0$ is contained in an open hemisphere. Then the solution μ to (1.3)-(1.2) with data μ_0 and $(\mathbf{V}(\cdot), \mathbf{B}(\cdot), \mathbf{W}(\cdot)) \equiv (I_d, \mathbf{B}, 0)$ satisfies $\operatorname{diam}(\operatorname{conv}_g \operatorname{supp} \mu(t)) \to 0$

as $t \to +\infty$. Moreover, for $\varepsilon > 0$ there exist $z \in \operatorname{conv}_g \operatorname{supp} \mu_0$ and $T > 0$ such that

$$W_\infty(\mu(T), \delta_z) \le \varepsilon \quad \text{and} \quad \inf\{t \ge 0 : W_2(\mu(t), \delta_z) \le \varepsilon\} = O\Big(\log \tfrac{1}{\varepsilon}\Big).$$

Proof sketch (full proof p. 11; an adaptation of [GLPR25, Lemma 6.4]). The characteristic flow is $\dot{x}(t) = \mathbf{P}^\perp_{x(t)} \mathcal{A}_{\mathbf{B}}[\mu(t)](x(t))$. Since $\operatorname{supp} \mu_0$ lies in an open hemisphere, the normalized attention vector $\gamma(x) := \mathcal{A}_{\mathbf{B}}[\mu_0](x)/\|\mathcal{A}_{\mathbf{B}}[\mu_0](x)\|$ is well defined on $\operatorname{conv}_g \operatorname{supp} \mu_0$ and points strictly into its interior. A first-order expansion at a boundary point x_0 gives $\langle x(\tau) - x_0, \gamma(x_0)\rangle > 0$ for small $\tau > 0$, so boundary points move inward and the geodesic convex hulls are strictly nested in time, $\operatorname{conv}_g \operatorname{supp} \mu(t_2) \subset \operatorname{int} \operatorname{conv}_g \operatorname{supp} \mu(t_1)$ for $t_1 < t_2$. Hence $\phi(t) := \operatorname{diam}(\operatorname{conv}_g \operatorname{supp} \mu(t))$ decreases to some $\ell \ge 0$; $\ell > 0$ would, by compactness, produce boundary points that fail to move inward, contradicting the strict interior-pointing, so $\ell = 0$. Nestedness and vanishing diameter yield a unique z with $\mu \rightharpoonup \delta_z$. Picking T with $\operatorname{diam}(\operatorname{conv}_g \operatorname{supp} \mu(T)) \le \varepsilon$ and the coupling $(\mathrm{Id}, \mathsf{T})_\# \mu(T)$, $\mathsf{T} \equiv z$, gives $W_p(\mu(T), \delta_z) \le \varepsilon$; the logarithmic rate follows from [CLPR25, Theorem 2.3].

Our formalization. open
The authors' strategy is a Lyapunov/LaSalle argument on the geodesic convex hull: with the pure self-attention drift ($\mathbf{W} = 0$, $\mathbf{V} = I_d$) the hull's diameter is strictly decreasing while its mass stays inside it, so the measure contracts to a single Dirac, the exponential contraction giving the $O(\log \frac{1}{\varepsilon})$ time. We state this type-correctly in Lean — the open hemisphere is `supportedIn μ {x | 0 < ⟪e, x⟫}` for a unit `e`, the target a `Measure.dirac z`, closeness measured in `W2` — as an open `sorry` stub. We do **not** claim a proof: the diameter monotonicity, the LaSalle/Hartman-Grobman convergence, and the rate live in the continuity-equation flow axiom. The conclusion is validated numerically by experiment **E2**.

```
/-- **Proposition 2.1** (clustering to a point). A measure supported in an open
↪ hemisphere can be
driven arbitrarily `W₂`-close to a Dirac mass. Stub (`sorry`): `math.open`. The rate
`O(log 1/ε)` and the LaSalle/Hartman-Grobman convergence are in the axiom layer. -/
theorem prop_2_1 (μ : Measure (Eucl d)) (T ε : ℝ) (hT : 0 < T) (hε : 0 < ε)
    (e : Eucl d) (he : ‖e‖ = 1) (hhemi : supportedIn μ {x | 0 < ⟪e, x⟫}) :
    ∃ (θ : Params d) (z : Eucl d), W2 (measureFlow θ T μ) (Measure.dirac z) ≤ ε := by
  sorry
```

Statements/MidLevel.lean

3.4 Clustering disjoint-support measures to atoms (Proposition 2.2)

From the original paper (Geshkovski, Rigollet, Ruiz-Balet, Proposition 2.2, pp. 11-12). Suppose μ_0^i has no atoms and $\operatorname{conv}_g \operatorname{supp} \mu_0^i \cap \operatorname{conv}_g \operatorname{supp} \mu_0^j = \emptyset$ for $i \ne j$. Fix $M \ge 1$, and for any $i \in [\![1, N]\!]$ consider

$$\mu_1^i := \sum_{k=1}^{M} \alpha_k^i \, \delta_{x_k^i} \in \mathcal{P}(\mathbb{S}^{d-1})$$

where $x_k^i \in \mathrm{conv}_g \operatorname{supp} \mu_0^i$, with $x_k^i = x_{k'}^j$ if and only if $(k,i) = (k',j)$. Then for any $T > 0$ and $\varepsilon > 0$ there exist piecewise constant $(\mathbf{W}, \mathbf{U}, b) : [0,T] \to \mathscr{M}_{d\times d}(\mathbb{R})^2 \times \mathbb{R}^d$ such that for any i, the corresponding solution μ^i to (1.3)-(1.2) with data μ_0^i, $\mathbf{V} \equiv 0$ and the above parameters, satisfies $W_2(\mu^i(T), \mu_1^i) \leq \varepsilon$ as well as, for $i \neq j$,

$$\mathrm{conv}_g \operatorname{supp} \mu^i(T) \cap \mathrm{conv}_g \operatorname{supp} \mu^j(T) = \emptyset.$$

The number of switches in $(\mathbf{W}, \mathbf{V}, b)$ can also be accounted for — see Remark 2.3.
Proof sketch (full proof from p. 12). Split $[0,T] = \bigcup_{i \in [\![1,N]\!]} [T_{i-1}, T_i]$ and apply Lemma B.1 separately within each interval, acting on one measure at a time. Property (B.2) — the flow is the identity off the active balls — ensures that while the construction acts on the i-th measure the others are untouched, $\mu^i(T_{i-1}) = \mu_0^i$. For a fixed i, partition the support $\mathcal{C}^i = \operatorname{supp} \mu_0^i$ into M pairwise disjoint pieces $\mathcal{C}_k^i$ with connected interiors, $\mu_0^i(\mathcal{C}_k^i) = \alpha_k^i$ and $x_k^i \in \operatorname{int} \mathcal{C}_k^i$; then steer the mass of each piece, through a chain of overlapping balls, into a small ball around x_k^i, so $W_2(\mu^i(T), \mu_1^i) \leq \varepsilon$ while the disjoint supports remain disjoint.

Our formalization. open
The authors' strategy chains the two preceding results: partition each atomless support into M prescribed-mass pieces, then use the ball-chain transport (Lemma B.1) to drive each piece into a small ball about its target atom x_k^i, handling the disjoint families one at a time so that — by the off-support identity (B.2) — the others stay fixed. We state the single-family conclusion type-correctly in Lean against the axiom layer: the target is a genuine M-atom combination `∑ k, α k • Measure.dirac (x k)`, atomlessness is `∀ x, μ {x} = 0`, and closeness is in `W2`. It is an open `sorry` stub — not a claimed proof — resting on the flow and optimal-transport axioms together with the (open) Lemma B.1 and Proposition 2.1.

```
/-- **Proposition 2.2** (clustering to a discrete measure). An atomless measure whose
↪ support lies in
a region `S` disjoint from the others can be driven `W₂`-close to a prescribed `M`-atom
↪ empirical
measure. Stub (`sorry`): `math.open`. -/
theorem prop_2_2 (μ ν_target : Measure (Eucl d)) (T ε : ℝ) (hT : 0 < T) (hε : 0 < ε)
    (hatomless : ∀ x : Eucl d, μ {x} = 0)
    (M : ℕ) (x : Fin M → Eucl d) (α : Fin M → ℝ≥0∞)
    (htgt : ν_target = ∑ k : Fin M, α k • Measure.dirac (x k)) :
    ∃ θ : Params d, W2 (measureFlow θ T μ) ν_target ≤ ε := by
  sorry
```

Statements/MidLevel.lean

4 Disentangling supports

A piecewise-constant control separates the N overlapping input supports into pairwise-disjoint regions of the sphere, by steering each measure's *barycenter* into its own direction and then shrinking its geodesic hull around it. This disentanglement is the heart of the construction behind Theorems 1.1 and 1.2: once the supports are disjoint, each measure can be clustered and matched independently.

From the original paper (Geshkovski, Rigollet, Ruiz-Balet, §3, p. 14).
"We show that flows generated by self-attention can disentangle measures with overlapping supports—it actually suffices to consider $\mathbf{B} \equiv 0$." Set $\mathbb{Q}_1^{d-1} := \mathbb{S}^{d-1} \cap (\mathbb{R}_{>0})^d$ and

$$\mathsf{v}[\mu](t,x) = \mathbf{P}_x^{\perp}(\mathbf{V}(t)\,\mathbb{E}_{\mu(t)}[z] + \mathbf{W}(t)(\mathbf{U}(t)x + b(t))_+). \tag{3.1}$$

The whole section uses this reduced field (3.1): a perceptron term $\mathbf{W}(\mathbf{U}x+b)_+$ and a barycentric attention term $\mathbf{V}\,\mathbb{E}_\mu[z]$, both projected onto the tangent space.

4.1 Transport to the orthant

From the original paper (Geshkovski, Rigollet, Ruiz-Balet, Lemma 3.2, p. 15).
Suppose $T > 0$ and $\mu_0^i \in \mathcal{P}(\mathbb{S}^{d-1})$ with $\bigcup_i \operatorname{supp} \mu_0^i \subsetneq \mathbb{S}^{d-1}$. There exists a piecewise constant $\mathbf{W} : [0,T] \to \mathcal{M}_{d\times d}(\mathbb{R})$, having at most one switch and satisfying

$$\|\mathbf{W}\|_{L^\infty((0,T);\mathcal{M}_{d\times d}(\mathbb{R}))} \leq C/T$$

for some $C = C(N) > 0$, such that for any $i \in [\![1,N]\!]$ the solution μ^i to (1.3)–(1.2) with data μ_0^i and $\mathbf{V} \equiv \mathbf{B} \equiv \mathbf{U} \equiv 0$, $b \equiv \mathbf{1}$, satisfies $\operatorname{supp} \mu^i(T) \subset \mathbb{Q}_1^{d-1}$.
Proof. *(proof sketch; full proof p. 15.)* Pick a direction $\omega \in \mathbb{S}^{d-1} \setminus \bigcup_i \operatorname{supp} \mu_0^i$ not filling the sphere and set $\mathbf{W}_1\mathbf{1} = -\omega$. The characteristics solve $\dot{x} = -\mathbf{P}_x^{\perp}\omega$, so $\frac{d}{dt}\langle x,\omega\rangle = -1+\langle x,\omega\rangle^2 < 0$ off $\{\pm\omega\}$, hence by some T_0 all supports lie in $B(-\omega,\pi/8)$. Pick $\alpha \in \mathbb{Q}_1^{d-1}$ with $d_g(-\omega,-\alpha) > \pi/8$ (so $-\alpha$ is off every support) and on $[T_0,T]$ take $\mathbf{W}_2\mathbf{1} = \alpha$: now $\frac{d}{dt}\langle x,\alpha\rangle = 1-\langle x,\alpha\rangle^2 \geq 0$ drives every point into a prescribed small cap around $\alpha \in \mathbb{Q}_1^{d-1}$. One switch; the norm bound follows by time-rescaling the two phases. □

Our formalization. open
The authors' construction (attributed to them) is the one switch above: a first phase $\mathbf{W}_1\mathbf{1} = -\omega$ contracts every support toward $-\omega$ through the gate-free dynamics $\frac{d}{dt}\langle x,\omega\rangle = -(1-\langle x,\omega\rangle^2)$ — the projector identity of leaf **L1** — and a second phase $\mathbf{W}_2\mathbf{1} = \alpha \in \mathbb{Q}_1^{d-1}$ drives them into a small cap inside the orthant. We give a **type-correct Lean statement against the axiom layer** (`measureFlow`, `switches`), expressing "support inside $\mathbb{Q}_1^{d-1}$" measure-theoretically as `μ (orthant d)ᶜ = 0`. It is a `sorry` stub: we state it, we do **not** claim a proof; its discharge rests on the continuity-equation flow axiom.

```
/-- The open positive orthant `Q₁^{d-1} = 𝕊^{d-1} ∩ (ℝ_{>0})^d`, as a subset of `ℝ^d`. -/
def orthant (d : ℕ) : Set (Eucl d) := {x | ∀ i, 0 < x i}

/-- "The support of `μ` is contained in `S`", expressed measure-theoretically as `μ(Sᶜ)
 ↪  = 0`. -/
def supportedIn (μ : Measure (Eucl d)) (S : Set (Eucl d)) : Prop := μ Sᶜ = 0

/-- **Lemma 3.2** (transport into the orthant). One parameter switch moves the measure
 ↪  into
`Q₁^{d-1}`. Stub (`sorry`): `math.open`. -/
theorem lemma_3_2 (μ : Measure (Eucl d)) (T : ℝ) (hT : 0 < T) :
    ∃ θ : Params d, switches θ ≤ 1 ∧ supportedIn (measureFlow θ T μ) (orthant d) := by
  sorry
```

Statements/MidLevel.lean

4.2 Shrinking a hull to its barycenter direction

From the original paper (Geshkovski, Rigollet, Ruiz-Balet, Lemma 3.3, p. 16; proof App. B.2, p. 33).
Let $\mu_0^i \in \mathscr{P}(\mathbb{Q}_1^{d-1})$ be such that $\mathbb{E}_{\mu_0^i}[x]$ is not colinear with $\mathbb{E}_{\mu_0^j}[x]$ for $i \neq j$. Then for any $T > 0$, $\varepsilon > 0$, and $\nu_0 \in \mathscr{P}(\mathbb{Q}_1^{d-1})$ such that $\mathbb{E}_{\nu_0}[x]$ is colinear with $\mathbb{E}_{\mu_0^j}[x]$, there exists a piecewise constant $\theta : [0,T] \to \Theta$ having at most $O(d \cdot N)$ switches such that

$$\operatorname{conv}_g \operatorname{supp} \nu(T) \cup \operatorname{conv}_g \operatorname{supp} \mu^N(T) \subset B\Big(\mathbb{E}_{\mu_0^j}[z] \Big/ \|\mathbb{E}_{\mu_0^j}[z]\|,\ \varepsilon\Big)$$

and $\mu^i(T) = \mu_0^i$ for $i \neq j$, where μ^i, ν denote the unique solutions to (1.3)–(3.1) corresponding to data μ_0^i, ν_0, and parameters θ.
Proof. *(proof sketch; full proof App. B.2, p. 33.)* Three steps. **(1) Isolating** μ_0^N, ν_0: with $\mathbf{W} \equiv 0$ and $\mathbf{V}(t) = \sum_k \alpha_k \alpha_k^\top \mathbf{1}_{[T_{k-1},T_k]}$, where $\{\alpha_k\}$ is an orthonormal basis of $\operatorname{span}(\mathbb{E}_{\mu_0^N}[z])^\perp$, the barycenter ODE (B.9) $\frac{d}{dt}\langle x, \alpha_1\rangle = \langle \mathbb{E}_{\mu(t)}[x], \alpha_1\rangle(1 - \langle \alpha_1, x\rangle^2)$ sends $x(t) \to \pm\alpha_k$, concentrating $\operatorname{supp}\mu(T_{d-1})$ near $\bigcup_k B(\pm\alpha_k, C_k\varepsilon_k) \subset \mathbb{S}^{d-1} \setminus \mathbb{Q}_1^{d-1}$ (flow Ψ_1, fixing μ_0^j for $j \leq N-1$). **(2) Clustering:** a separating half-space $\langle a, x\rangle + \underline{b}$ with $\mathbf{U} = \mathbf{1}a^\top$, $\mathbf{W}_2\mathbf{1} = \mathbb{E}_{\mu_0^N}[x]$ drives both supports into $B(\mathbb{E}_{\mu_0^N}[x]/\|\mathbb{E}_{\mu_0^N}[x]\|, \delta)$ (flow Ψ_2). **(3) Flow reversal:** $\Phi_{\text{fin}} := \Psi_1^{-1} \circ \Psi_2 \circ \Psi_1$ returns μ_0^i to itself for $i \leq N-1$ and clusters ν, μ^N within $B(\,\cdot\,, C_T\delta)$; pick δ small. ☐

Our formalization. open
The authors' three-step argument (App. B.2) hinges on the **barycenter ODE (B.9)**, which is exactly our machine-checked leaf **L6** (`barycenter_hasDerivAt_inner`): along the attention field, $\frac{d}{dt}\langle x, \alpha\rangle = c\,(1 - \langle x, \alpha\rangle^2)$ with strict increase for $c > 0$, which is what drives each point to $\pm\alpha$ and lets the support be concentrated into an ε-ball around the barycenter direction. We state the conclusion type-correctly against the axiom layer: some θ concentrates the flow into `Metric.ball α ε`. It is a `sorry` stub stated against `measureFlow`; the analytic content (the three flows $\Psi_1, \Psi_2, \Psi_1^{-1}$) lives in the flow axiom.

```
/-- **Lemma 3.3** (shrink a measure's hull toward its barycenter direction). For any
 ↪  tolerance the
measure can be concentrated into a small ball around some direction `α`, with `O(dN)`
 ↪  switches.
Stub (`sorry`): `math.open`. -/
theorem lemma_3_3 (μ : Measure (Eucl d)) (T ε : ℝ) (hT : 0 < T) (hε : 0 < ε) :
    ∃ (θ : Params d) (α : Eucl d), supportedIn (measureFlow θ T μ) (Metric.ball α ε) :=
 ↪  by
  sorry
```

Statements/MidLevel.lean

4.3 Making barycenters non-colinear

From the original paper (Geshkovski, Rigollet, Ruiz-Balet, Lemma 3.4, p. 16; proof App. B.3, p. 35).
Let $T > 0$ and let $\mu_0, \nu_0 \in \mathcal{P}(\mathbb{Q}_1^{d-1})$ be two different measures such that $\mathbb{E}_{\mu_0}[x] = \gamma_1 \mathbb{E}_{\nu_0}[x]$ for some $\gamma_1 \in (0, 1]$.

1. If $\gamma_1 = 1$, then, setting $\mathbf{V} \equiv 0$, there exist $\mathbf{W}, \mathbf{U} \in \mathscr{M}_{d\times d}(\mathbb{R})$ and $b \in \mathbb{R}^d$ such that the solutions μ, ν to (1.3)–(3.1) corresponding to μ_0, ν_0 and these parameters, satisfy $\mathbb{E}_{\mu(T)}[x] \neq \mathbb{E}_{\nu(T)}[x]$. Moreover the Lipschitz-continuous and invertible flow map $\Phi^T : \mathbb{S}^{d-1} \to \mathbb{S}^{d-1}$ induced by the characteristics satisfies $\Phi^T(x) = x$ for $x \in \mathbb{S}^{d-1} \setminus (\operatorname{conv}_g \operatorname{supp} \mu_0 \cup \operatorname{conv}_g \operatorname{supp} \nu_0)$. (3.2)
2. If $\gamma_1 \neq 1$, then, setting $\mathbf{B} \equiv 0$, there exist piecewise-constant $(\mathbf{V}, \mathbf{W}, \mathbf{U})$ and b with at most 2 switches such that $\mathbb{E}_{\mu(T)}[x] \neq \gamma_2\, \mathbb{E}_{\nu(T)}[x]$ for all $\gamma_2 \in \mathbb{R}$ (non-colinear).

Proof (Part 1). *(proof sketch; full proof App. B.3, p. 35.)* Since $\mu_0 \neq \nu_0$, there is an open ball $\mathcal{B} \subset \operatorname{supp} \mu_0 \cup \operatorname{supp} \nu_0$ with $\mu_0(\mathcal{B}) \neq \nu_0(\mathcal{B})$. There exists $x^* \in \mathcal{B}$ with

$$\mu_0(\mathcal{B})x^* + \int_{\mathbb{S}^{d-1}\setminus\mathcal{B}} x\, \mu_0(\mathrm{d}x) \;\neq\; \nu_0(\mathcal{B})x^* + \int_{\mathbb{S}^{d-1}\setminus\mathcal{B}} x\, \nu_0(\mathrm{d}x),$$

for otherwise $x^* = \frac{1}{\mu_0(\mathcal{B})-\nu_0(\mathcal{B})} \int_{\mathbb{S}^{d-1}\setminus\mathcal{B}} x(\nu_0 - \mu_0)(\mathrm{d}x)$ would be forced constant over all of $\mathcal{B}$ — impossible. Taking such x^*, a perceptron block $\mathbf{U} = -\mathbf{1}a^\top$, $b = R\mathbf{1}$ (a, R the center and radius of $\mathcal{B}$), and $\mathbf{W1} = x^*$, by Lemma B.2 pushes $\mu(T)$ to $\mu_0(\mathcal{B})\delta_{x^*} + \mu_0(\cdot \setminus \mathcal{B})$; continuity of the expectation in W_2 yields $\mathbb{E}_{\Phi_\# \mu_0}[x] \neq \mathbb{E}_{\Phi_\# \nu_0}[x]$, with $\Phi(x) = x$ off $\mathcal{B}$. □

Our formalization. open
In the $\gamma_1 = 1$ (perceptron) case the authors locate a ball $\mathcal{B}$ on which μ_0, ν_0 differ in mass, then argue that **some** $x^* \in \mathcal{B}$ **makes the barycenters differ** — for if every x^* gave equality, x^* would be pinned to one value on the whole ball, which no map can be. That "no map is constant on a nonempty open ball" core is our machine-checked leaf **L10** (`exists_ne_in_ball`). We state the $\gamma_1 = 1$ conclusion type-correctly: equal input barycenters force the existence of

a θ whose output barycenters differ. `sorry` stub against the axiom layer. (The attention case $\gamma_1 \neq 1$, at most two switches, is the companion stub `lemma_3_4_part2`.)

```
/-- **Lemma 3.4, Part 1** (`γ₁ = 1` case). If two measures have equal barycenters, a
↪ constant
parameter makes the barycenters differ. Stub (`sorry`): `math.open`. The self-contained
↪ pigeonhole
core is leaf L10. -/
theorem lemma_3_4_part1 (μ ν : Measure (Eucl d)) (T : ℝ) (hT : 0 < T)
    (hbar : barycenter μ = barycenter ν) :
   ∃ θ : Params d, barycenter (measureFlow θ T μ) ≠ barycenter (measureFlow θ T ν) := by
  sorry
```

Statements/MidLevel.lean

4.4 Disentanglement

From the original paper (Geshkovski, Rigollet, Ruiz-Balet, Proposition 3.1, pp. 15-17). Let $T > 0$ and $\mu_0^i \in \mathcal{P}(\mathbb{Q}_1^{d-1})$. There exists a piecewise constant $\theta : [0,T] \to \Theta$, having at most $O(d \cdot N)$ switches, such that for all $i \in [\![1, N]\!]$, the solution μ^i to (1.3)–(3.1) with data μ_0^i and θ satisfies

$$\mathrm{conv}_g \operatorname{supp} \mu^i(T) \cap \mathrm{conv}_g \operatorname{supp} \mu^j(T) = \emptyset \quad \text{if } i \neq j.$$

Proof. *(proof sketch; full proof pp. 16-17.)* Induction on N; the base case $N = 1$ is trivial. Assume the first $N-1$ measures already have pairwise-disjoint geodesic hulls (3.3) and let μ_0^N be arbitrary. Since $\operatorname{supp} \mu_0^i \subset \mathbb{Q}_1^{d-1}$, (3.3) implies $\mathbb{E}_{\mu_0^i}[x]$ is not colinear with $\mathbb{E}_{\mu_0^j}[x]$ for $i \neq j \leq N-1$. If $\mathbb{E}_{\mu_0^N}[x]$ is non-colinear with all of them, one application of Lemma 3.3 (small ε) shrinks μ^N's hull until it separates. Otherwise $\mathbb{E}_{\mu_0^N}[x]$ is colinear with $\mathbb{E}_{\mu_0^i}[x]$ for at most one i (WLOG $i = N-1$), and four phases of $[0,T]$ finish it: (1) $[0, T/4]$ Lemma 3.3 peels μ^N, μ^{N-1} off the rest; (2) $[T/4, T/2]$ Lemma 3.4 Part 1 makes $\mathbb{E}_{\mu^{N-1}}[x] \neq \mathbb{E}_{\mu^N}[x]$ while (3.2) keeps the others fixed; (3) $[T/2, 3T/4]$ Lemma 3.4 Part 2 makes the two barycenters non-colinear; (4) $[3T/4, T]$ Lemma 3.3, now applicable to all N measures, separates them. Total $O(d \cdot N)$ switches. $\square$

Our formalization. open
We state the headline controllability conclusion of the authors' induction type-correctly against the axiom layer: there is a θ whose solution map renders the family's supports pairwise disjoint, recorded by the opaque predicate `DisjointSupports`. It is a `sorry` stub — the four-phase induction, the $O(dN)$ switch count, and the geodesic-hull nesting all rest on the flow / conv_g axioms, so we do not claim a proof.

```
/-- The measures in the family have pairwise disjoint (geodesic-convex-hull) supports.
 ↪  -/
axiom DisjointSupports {N : ℕ} (ν : Fin N → Measure (Eucl d)) : Prop

/-- **Proposition 3.1** (disentanglement). There is a piecewise-constant `θ` whose
 ↪  solution map
renders the supports of the family pairwise disjoint. Stub (`sorry`): status
 ↪  `math.open`. -/
theorem prop_3_1 (hd : 3 ≤ d) {N : ℕ} (μ₀ : Fin N → Measure (Eucl d)) (T : ℝ) (hT : 0 <
 ↪  T) :
    ∃ θ : Params d, DisjointSupports (fun i ⇒ measureFlow θ T (μ₀ i)) := by
  sorry
```

Statements/MainResults.lean

What we strengthen. *(finding F2.)* The induction above opens by asserting that **disjoint geodesic hulls imply non-colinear barycenters** —

$$\operatorname{conv}_g \operatorname{supp} \mu_0^i \cap \operatorname{conv}_g \operatorname{supp} \mu_0^j = \emptyset \implies \mathbb{E}_{\mu_0^i}[x] \text{ not colinear with } \mathbb{E}_{\mu_0^j}[x]$$

— **without proof**. Our adversarial review flagged this as the one genuine rigor gap (finding F2). It is now the machine-checked leaf **L11** (`barycenter_noncolinear_of_disjoint_hull`, see the Self-contained leaves section) for the empirical regime of Theorem 1.1 / restricted Theorem 1.2, and **L11**′ (`barycenter_noncolinear_of_disjoint_hull_general`) for arbitrary probability measures via Mathlib's `Convex.integral_mem`. The barycenter, a nonnegative average of support points, lies in cone(supp); colinearity would put a common normalized direction in *both* hulls, contradicting disjointness. The headline `prop_3_1` stays `[open]{.badge .open}` only because it still rests on the flow / conv_g-nesting axioms — not on this step. (Validated numerically by experiment E3.)
Leaves/BarycenterNonColinear.lean

5 Matching discrete measures

Once the disentanglement of §3 has rendered the supports pairwise disjoint and the clustering of §2 has collapsed each measure to a single point, matching reduces to steering finitely many points to their prescribed targets. A single perceptron gate — identically zero on a chosen parking cap and positive outside it — lets the flow drive one *active* point along a geodesic corridor to its target while every other point, parked inside the cap, stays fixed.

5.1 Proposition 4.2 — steer the active point

From the original paper (Geshkovski, Rigollet, Ruiz-Balet, §4, Proposition 4.2, p. 18). Suppose $d \geqslant 3$. Consider

$$(x_0^i, y^i) \in \mathbb{S}^{d-1} \times \mathbb{S}^{d-1} \qquad \text{for } i \in [\![1, M]\!], \tag{$\mathcal{D}$}$$

with $x_0^i \neq x_0^j$ and $y^i \neq y^j$ for $i \neq j$, with $x_0^i = y^i$ for $i \in [\![1, M-1]\!]$, and suppose that there exist $\gamma \in \mathbb{S}^{d-1}$ and $\varepsilon > 0$ such that $\langle \gamma, x_0^M - y^M \rangle = 0$ and $x_0^i \notin H_\varepsilon^\gamma$ for all $i \in [\![1, M-1]\!]$, where $H_\varepsilon^\gamma := \{x \in \mathbb{S}^{d-1} : |\langle x, \gamma \rangle| \leqslant \varepsilon\}$.
For any $T > 0$ there exist piecewise constant $\theta = (\mathbf{W}, \mathbf{U}, b) : [0, T] \to \mathcal{M}_{d\times d}(\mathbb{R})^2 \times \mathbb{R}^d$, having at most 6 switches, such that for any i, the solution $x^i(\cdot)$ to

$$\begin{cases} \dot{x}^i(t) = \mathbf{P}_x^\perp \mathbf{W}(t)\big(\mathbf{U}(t)x^i(t) + b(t)\big)_+ & \text{in } [0, T] \\ x^i(0) = x_0^i, \end{cases} \tag{4.1}$$

satisfies $x^i(T) = y^i$. Moreover, there exists $C > 0$, not depending on $\mathcal{D}$ and T, such that

$$\|\theta\|_{L^\infty((0,T);\Theta)} \leqslant \frac{C}{T \cdot \varepsilon}.$$

Proof sketch (proof sketch; full proof pp. 18-23). The control is a 6-interval step function $(\mathbf{W}(t), \mathbf{U}(t), b(t)) = \sum_{j=1}^6 (\mathbf{W}_j, \mathbf{U}_j, b_j)\, \mathbf{1}_{[(j-1)T/6,\, jT/6]}(t)$ with $\mathbf{U}_5 = \mathbf{U}_1$, $\mathbf{U}_6 = \mathbf{U}_2$, $\mathbf{U}_3 = \mathbf{U}_4$, $b_5 = b_1$, $b_6 = b_2$, $b_3 = b_4$, $\mathbf{W}_5 = -\mathbf{W}_1$, $\mathbf{W}_6 = -\mathbf{W}_2$, i.e. the last two intervals time-reverse the first two — at most 6 switches. *Anchors (Step 1).* Since $\langle \gamma, x_0^M - y^M \rangle = 0$ one finds $\omega \in \mathbb{S}^{d-1}$ with $\langle \gamma, \omega \rangle = 0$ and $d_g(\omega, x_0^M), d_g(\omega, y^M) \geqslant \pi/2$; the points $\omega_\pm$ at geodesic distance $\pi/8$ from ω toward $\pm\gamma$ then satisfy $d_g(\omega_\pm, x_0^M), d_g(\omega_\pm, y^M) \geqslant 3\pi/8$, so $\{\langle \omega, x \rangle = \cos(\pi/8 + \tau)\}$ separates the cap $B(\omega, \pi/8 + \tau)$ from x_0^M, y^M. *Phase 1 — "gather" ψ_1 (Steps 1-2, 2 switches).* Two opposite gates $(\langle \pm\gamma, x \rangle - \epsilon/2)_+$ with drift targets $\omega_\pm$ run the Riemannian gradient flow of $d_g(\cdot, \omega_\pm)$ (using $\nabla_1 d_g(x, \omega_+) = -\mathbf{P}_x^\perp \omega_+ / \sqrt{1 - \langle x, \omega_+ \rangle^2}$); $\langle x, \omega_\pm \rangle$ increases monotonically (4.5) and the trajectory settles exponentially near $\omega_\pm$ by LaSalle / Hartman-Grobman (4.6), so after time $T/3$ every inactive point lies in the cap $B(\omega, 3\pi/16)$ while points of H_ε^γ stay fixed (4.7). *Phase 2 — "corridor" ψ_2 (Step 3, 2 switches).* The gate $(\langle -\omega, x \rangle + \cos(3\pi/16))_+$ is identically 0 on $B(\omega, 3\pi/16)$ and positive outside; driving x_0^M first toward an intermediate z_1 then toward z_2 along a geodesic corridor that lies outside the cap and passes through y^M lands the active point at y^M, while the inactive points do not move (gate off on the cap). *Phase 3 — "restore" ψ_1^{-1} (Step 4, 2 switches).* Re-running the gather action with flipped drifts time-reverses Phase 1, returning every inactive point to its start; the active point remains at y^M since the corridor gate is off on the cap. The composition $(\psi_1)^{-1} \circ \psi_2 \circ \psi_1$ sends $x_0^M \mapsto y^M$ and fixes all other x_0^i; choosing drift size $\|\mathbf{W}\| \asymp (T\varepsilon)^{-1}$ gives total time T and $\|\theta\|_{L^\infty} \lesssim 1/(T\varepsilon)$.

Our formalization. open
The authors' construction (pp. 18-23) realises the matching of one point as the gather / corridor / restore composition $(\psi_1)^{-1} \circ \psi_2 \circ \psi_1$, a 6-interval piecewise-constant control whose middle phase steers the active x_0^M to y^M while a gate parked on the cap $B(\omega, 3\pi/16)$ freezes the inactive points. We record a type-correct Lean **statement** of this proposition as a `sorry`

(open): with $d \geqslant 3$, the inactive points already at their targets (`hfix`), at most 6 switches drive every input to its target while keeping the inactive ones fixed. We do **not** claim a proof. Two of the construction's cores are however discharged as kernel-checked leaves: the separating-hyperplane estimate of Step 1 is leaf **L3** (`Leaves/SeparatingHyperplane.lean`, `separating_hyperplane`), and the geodesic-gradient identity driving the gather and corridor flows is leaf **L4** (`Leaves/GeodesicGradient.lean`, `geodesicDist_hasDerivAt`). The selective-motion mechanism — active point reaches the target, parked points stay fixed — is validated numerically by experiment **E4**.

```
/-- **Proposition 4.2** (steer one active point). With `d ≥ 3` and the inactive points
↪  already at
their targets, at most `6` switches move every input to its target, keeping the inactive
↪  ones fixed
(`x₀ i = y i` is preserved by `flowMap θ T (x₀ i) = y i`). Stub (`sorry`): `math.open`.
↪  Step 1 is
leaf L3, the geodesic gradient is leaf L4. -/
theorem prop_4_2 (hd : 3 ≤ d) (M : ℕ) (x₀ y : Fin M → Eucl d) (T : ℝ) (hT : 0 < T)
    (hfix : ∀ i : Fin M, (i : ℕ) < M - 1 → x₀ i = y i) :
    ∃ θ : Params d, switches θ ≤ 6 ∧ ∀ i, flowMap θ T (x₀ i) = y i := by
  sorry
```

`Statements/MidLevel.lean`

5.2 Proposition 4.1 — match the ensemble

From the original paper (Geshkovski, Rigollet, Ruiz-Balet, §4, Proposition 4.1, p. 18). Suppose $d \geqslant 3$. Consider

$$(x_0^i, y^i) \in \mathbb{S}^{d-1} \times \mathbb{S}^{d-1} \qquad \text{for } i \in [\![1, M]\!], \tag{$\mathcal{D}$}$$

with $x_0^i \neq x_0^j$ and $y^i \neq y^j$ for $i \neq j$, and suppose that for any i there exist $\gamma_i \in \mathbb{S}^{d-1}$ and $\varepsilon_i > 0$ such that $\langle \gamma_i, x_0^i - y^i \rangle = 0$ and $x_0^j \notin H_{\varepsilon_i}^{\gamma_i}$ for $j \neq i$, where

$$H_{\varepsilon_i}^{\gamma_i} := \{x \in \mathbb{S}^{d-1} : |\langle x, \gamma_i \rangle| \leqslant \varepsilon_i\}.$$

For any $T > 0$ there exist piecewise constant $\theta = (\mathbf{W}, \mathbf{U}, b) : [0, T] \to \mathcal{M}_{d\times d}(\mathbb{R})^2 \times \mathbb{R}^d$, having at most $6M$ switches, such that for any i, the solution $x^i(\cdot) \in \mathcal{C}^0([0, T]; \mathbb{S}^{d-1})$ to (4.1) satisfies $x^i(T) = y^i$. Moreover, there exists $C > 0$, not depending on $\mathcal{D}$ nor T, such that

$$\|\theta\|_{L^\infty((0,T);\Theta)} \leqslant \frac{C \cdot M}{T \, \min_i \varepsilon_i}.$$

Proof sketch (proof sketch; full proof p. 18). "The proof of Proposition 4.1 follows directly from [Proposition 4.2], combined with a straightforward induction argument": activating one pair (x_0^i, y^i) at a time, each application of Proposition 4.2 steers the i-th point to its target in at most 6 switches while leaving the already-matched points fixed, so M applications cost at most $6M$ switches and the norm bound accumulates to $C \cdot M/(T \min_i \varepsilon_i)$.

Our formalization. open

The authors obtain Proposition 4.1 by induction over the M pairs, each step being one application of Proposition 4.2 (the active point reaches its target; the previously matched points are inactive and stay fixed), giving the $\leqslant 6M$-switch budget and the $O(CM/(T\min_i \varepsilon_i))$ norm. We record a type-correct Lean **statement** as a `sorry` (open): with $d \geqslant 3$ and injective inputs/targets, at most $6M$ switches steer every x_0^i to y^i. We do **not** claim a proof; the discharge would be the M-fold composition of `prop_4_2`. (Validated numerically by experiment **E4**, alongside Proposition 4.2.)

```
/-- **Proposition 4.1** (match an ensemble). With `d ≥ 3` and distinct inputs/targets,
 ↪  at most `6M`
switches steer every `x₀ i` to `y i`. Stub (`sorry`): `math.open`. Follows from
 ↪  Proposition 4.2 by
induction. -/
theorem prop_4_1 (hd : 3 ≤ d) (M : ℕ) (x₀ y : Fin M → Eucl d) (T : ℝ) (hT : 0 < T)
    (hx₀ : Function.Injective x₀) (hy : Function.Injective y) :
    ∃ θ : Params d, switches θ ≤ 6 * M ∧ ∀ i, flowMap θ T (x₀ i) = y i := by
  sorry
```

Statements/MidLevel.lean

6 Main results

The paper assembles its leaves and mid-level propositions into one solution map Φ^T_{fin} by a three-stage recipe: **disentangle** the N supports until they are pairwise disjoint (Proposition 3.1), **cluster** each measure — to a single point (Proposition 2.1) or to an M-atom empirical measure (Proposition 2.2) — and **match** each cluster to its prescribed target (Propositions 4.1/4.2). Theorem 1.1 is the Dirac-target case, steering each input to a point mass δ_{x^i} with $O(d\,N)$ switches and an explicit norm bound, while Theorem 1.2 is the general case, reaching any L^2 transport-matchable targets by interposing the map-transfer Lemma 5.1, the flow-approximation Lemma 5.4, and the coupling bound Lemma 5.2. Every node here is stated type-correctly in Lean against the axiom layer and left as a `sorry` (status open), except Proposition 6.2 — the generic $O(1)$-switch disentanglement heuristic — which we reviewed on paper only.

6.1 Lemma 5.1 — transport maps preserved by disentanglement

From the original paper (Geshkovski, Rigollet, Ruiz-Balet, Lemma 5.1, p. 24). Suppose that for every i there exists $\mathsf{T}^i \in L^2(\mathbb{S}^{d-1};\mathbb{S}^{d-1})$ with $\mathsf{T}^i_{\#}\mu_0^i = \mu_1^i$. Consider the flow map $\Phi_1 : \mathcal{P}(\mathbb{S}^{d-1}) \to \mathcal{P}(\mathbb{S}^{d-1})$ (resp. Φ_3) given by Proposition 3.1 with data $\mu_0^i \in \mathcal{P}(\mathbb{S}^{d-1})$ (resp. $\mu_1^i \in \mathcal{P}(\mathbb{S}^{d-1})$). Then there exists a Lipschitz-continuous and invertible map $\psi : \mathbb{S}^{d-1} \to \mathbb{S}^{d-1}$ such that

$$\psi\big|_{\mathrm{supp}\,\Phi_1(\mu_0^i)} = \psi^i\big|_{\mathrm{supp}\,\Phi_1(\mu_0^i)} \tag{5.1}$$

for any $i \in \{1, \dots, N\}$, where $\psi^i : \mathbb{S}^{d-1} \to \mathbb{S}^{d-1}$ is another Lipschitz-continuous and invertible map that satisfies

$$(\psi^i \circ \mathsf{T}^i_{\Phi_1})_{\#}\mu^i_0 = (\psi \circ \mathsf{T}^i_{\Phi_1})_{\#}\mu^i_0 = (\mathsf{T}^i_{\Phi_3})_{\#}\mu^i_1 = \Phi_3(\mu^i_1)$$

for some Lipschitz-continuous and invertible maps $\mathsf{T}^i_{\Phi_1}, \mathsf{T}^i_{\Phi_3} : \mathbb{S}^{d-1} \to \mathbb{S}^{d-1}$. The proof can be found in Appendix B.4.

Our formalization. open
The paper's strategy (Geshkovski, Rigollet, Ruiz-Balet, App. B.4, p. 37) exploits disentanglement: once Proposition 3.1 has pushed the supports $\operatorname{supp} \Phi_1(\mu^i_0)$ pairwise disjoint, the per-pair transport maps T^i are carried through the flow as Lipschitz-invertible maps $\mathsf{T}^i_{\Phi_1}, \mathsf{T}^i_{\Phi_3}$, and the individual matchings ψ^i can be glued into one global Lipschitz-continuous, invertible ψ that agrees with each ψ^i on the i-th (now disjoint) support and realizes $(\psi \circ \mathsf{T}^i_{\Phi_1})_{\#}\mu^i_0 = \Phi_3(\mu^i_1)$. We give a type-correct Lean **statement** as a `sorry` (open), abstracting the Lipschitz/geodesic bookkeeping into the matchability hypothesis: from a per-index family of transport maps we ask for a single `Bijective` ψ pushing every μ^i_0 to μ^i_1. We do not claim a proof.

```
/-- **Lemma 5.1** (transport map after disentanglement). If each disentangled pair is
↪ matchable, a
single bijective map matches them all. Stub (`sorry`): `math.open`. -/
theorem lemma_5_1 {N : ℕ} (μ₀ μ₁ : Fin N → Measure (Eucl d))
    (hmatch : ∀ i, ∃ Ti : Eucl d → Eucl d, (μ₀ i).map Ti = μ₁ i) :
    ∃ ψ : Eucl d → Eucl d, Function.Bijective ψ ∧ ∀ i, (μ₀ i).map ψ = μ₁ i := by
  sorry
```

Statements/MidLevel.lean

6.2 Lemma 5.4 — L² approximation by a flow map

From the original paper (Geshkovski, Rigollet, Ruiz-Balet, Lemma 5.4, p. 24).
Suppose $\varepsilon > 0$ and $\mu \in \mathcal{P}(\mathbb{S}^{d-1})$. For every $\psi \in L^2(\mathbb{S}^{d-1}; \mathbb{S}^{d-1})$, there exists a Lipschitz-continuous and invertible map $\psi_\varepsilon : \mathbb{S}^{d-1} \to \mathbb{S}^{d-1}$ induced by the solution map of (B.1), namely $\Phi^T_{\theta_\varepsilon}(\mu) = (\psi_\varepsilon)_{\#}\mu$ for some piecewise constant $\theta_\varepsilon : [0, T] \to \Theta$ with finitely many switches, such that

$$\|\psi - \psi_\varepsilon\|_{L^2(\mu)} \leq \varepsilon.$$

The proof of Lemma 5.4 is involved, so we postpone it to Appendix B.5.

Our formalization. open
The paper (Geshkovski, Rigollet, Ruiz-Balet, App. B.5, p. 38) shows any $\psi \in L^2(\mathbb{S}^{d-1}; \mathbb{S}^{d-1})$ is approximated in $L^2(\mu)$, to any tolerance, by a flow map ψ_ε of the continuity equation (B.1) with piecewise-constant control and finitely many switches; chained with the coupling bound $W_2(\mathsf{T}^1_{\#}\mu, \mathsf{T}^2_{\#}\mu) \leq \|\mathsf{T}^1 - \mathsf{T}^2\|_{L^2(\mu)}$ (Lemma 5.2, leaf L7) this converts L^2 control of the map into W_2 control of the pushforward. We give a type-correct Lean **statement** as a `sorry` (open):

for any ψ and ε, a parameter θ and a map ψ_ε with `measureFlow θ T μ = μ.map ψε` and $\sqrt{\int \|\psi - \psi_\varepsilon\|^2 \, \mathrm{d}\mu} \le \varepsilon$. We do not claim a proof.

```
/-- **Lemma 5.4** (`L²` approximation by a flow map). Any transport map `ψ` is
↪  approximated in
`L²(μ)` by a flow map of the dynamics, to any tolerance, with finitely many switches.
↪  Stub
(`sorry`): `math.open`. Combined with the coupling bound (leaf L7) this controls `W₂`.
↪  -/
theorem lemma_5_4 (μ : Measure (Eucl d)) (ψ : Eucl d → Eucl d) (T ε : ℝ) (hT : 0 < T)
↪  (hε : 0 < ε) :
    ∃ (θ : Params d) (ψε : Eucl d → Eucl d),
      measureFlow θ T μ = μ.map ψε ∧ Real.sqrt (∫ x, ‖ψ x - ψε x‖ ^ 2 ∂μ) ≤ ε := by
  sorry
```

Statements/MidLevel.lean

6.3 Theorem 1.2 — matching transport-matchable targets

From the original paper (Geshkovski, Rigollet, Ruiz-Balet, Theorem 1.2, p. 5). Suppose $d \ge 3$. Consider data $(\mathcal{D})$ such that

(1) there exist $w_0, w_1 \in \mathbb{S}^{d-1}$ such that

$$w_0 \notin \bigcup_i \mathrm{supp}(\mu_0^i) \quad \text{and} \quad w_1 \notin \bigcup_i \mathrm{supp}(\mu_1^i); \tag{1.5}$$

(2) for any $i \in \{1, \dots, N\}$, there exists $\mathsf{T}^i \in L^2(\mathbb{S}^{d-1}; \mathbb{S}^{d-1})$ such that $\mathsf{T}^i_{\#}\mu_0^i = \mu_1^i$.

Then for any $T > 0$ and $\varepsilon > 0$, there exists $\theta \in L^\infty((0,T);\Theta)$ such that for any $i \in \{1, \dots, N\}$, the unique solution $\mu^i \in \mathcal{C}^0([0,T]; \mathcal{P}(\mathbb{S}^{d-1}))$ to (1.3) with data μ_0^i and parameters θ satisfies

$$\mathsf{W}_2(\mu^i(T), \mu_1^i) \le \varepsilon.$$

Moreover, θ can be chosen piecewise constant.
Proof sketch (full proof §5.1, p. 25). The construction $\Phi^T_{\mathrm{fin}} := (\Phi^{T/3}_{\theta_3})^{-1} \circ \Phi^{T/3}_{\theta_2} \circ \Phi^{T/3}_{\theta_1}$ proceeds in three steps. **(1) Disentanglement:** Proposition 3.1 renders the input supports (μ_0^i) pairwise disjoint via Φ_{θ_1} on $[0, T/3]$ and the target supports (μ_1^i) via Φ_{θ_3} on $[2T/3, T]$; well-posedness and time-reversibility give a constant C with $\mathsf{W}_2((\Phi^T_{\theta_3})^{-1}\mu, (\Phi^T_{\theta_3})^{-1}\nu) \le C\, \mathsf{W}_2(\mu,\nu)$ (5.3). **(2) Matching:** Lemma 5.1 produces a single Lipschitz-invertible ψ agreeing with each ψ^i on the disentangled support, and Lemma 5.4 realizes ψ up to L^2 error by a flow map $\psi_\varepsilon = \Phi_{\theta_2}$ on $[T/3, 2T/3]$; the coupling bound (Lemma 5.2) turns the L^2 error into $\mathsf{W}_2((\Phi^{2T/3}_{\theta_2} \circ \Phi^{T/3}_{\theta_1})(\mu_0^i), \Phi^T_{\theta_3}(\mu_1^i)) \le \varepsilon / C$ (5.5–5.6). **(3) Composition:** applying $(\Phi^T_{\theta_3})^{-1}$ and (5.3) yields $\mathsf{W}_2(\mu^i(T), \mu_1^i) \le \varepsilon$ for all i.

Our formalization. open

The paper's proof (Geshkovski, Rigollet, Ruiz-Balet, §5.1) composes the three stages above: disentangle both families by Proposition 3.1, transfer the transport maps into a single ψ by Lemma 5.1, approximate ψ by a flow map by Lemma 5.4, and bound the resulting W_2 error by the coupling Lemma 5.2. We give a type-correct Lean **statement** as a `sorry` (open). The hypothesis (1.5) — a shared missing direction off every input support and off every target support — and the L^2-matchability of the pairs are recorded as the opaque predicates `SharedMissingDirection` and `Matchable`, so the statement stays faithful without re-deriving the absent support / optimal-transport API; the conclusion is the controllability core, a single piecewise-constant θ with `W2 (measureFlow θ T (μ₀ i)) (μ₁ i) ≤ ε` for every i. We do not claim a proof.

```
/-- **Theorem 1.2** (general targets). If every input/target pair is matchable by a
 ↪  transport map
and the families share a missing direction, then for any horizon `T` and tolerance `ε`
 ↪  there is a
piecewise-constant parameter `θ` whose solution map steers each input measure to within
 ↪  `ε` of its
target in `W₂`. Stub (`sorry`): status `math.open`. -/
theorem theorem_1_2 (hd : 3 ≤ d) {N : ℕ} (μ₀ μ₁ : Fin N → Measure (Eucl d))
    (T ε : ℝ) (hT : 0 < T) (hε : 0 < ε)
    (hmiss₀ : SharedMissingDirection μ₀) (hmiss₁ : SharedMissingDirection μ₁)
    (hmatch : Matchable μ₀ μ₁) :
    ∃ θ : Params d, ∀ i, W2 (measureFlow θ T (μ₀ i)) (μ₁ i) ≤ ε := by
  sorry
```

Statements/MainResults.lean

6.4 Theorem 1.1 — matching Dirac targets

From the original paper (Geshkovski, Rigollet, Ruiz-Balet, Theorem 1.1, p. 5). Suppose $d \geq 3$. Consider data $(\mathcal{D})$ such that

(1) there exists $w_0 \in \mathbb{S}^{d-1}$ such that

$$w_0 \notin \bigcup_i \operatorname{supp}(\mu_0^i); \tag{1.4}$$

(2) for any $i \in \{1, \dots, N\}$, we have $\mu_1^i = \delta_{x^i}$.

Then for any $T > 0$ and $\varepsilon > 0$, there exists $\theta \in L^\infty((0,T);\Theta)$ such that for any $i \in \{1, \dots, N\}$, the unique solution $\mu^i \in \mathcal{C}^0([0,T];\mathcal{P}(\mathbb{S}^{d-1}))$ to (1.3) with data μ_0^i and parameters θ satisfies

$$\mathsf{W}_2(\mu^i(T), \mu_1^i) \leq \varepsilon.$$

Moreover, θ can be chosen piecewise constant, with $O(d \cdot N)$ switches, and

$$\|\theta\|_{L^\infty((0,T);\Theta)} = O\Big(\frac{d \cdot N}{T} + \log\frac{1}{\varepsilon}\Big).$$

Proof sketch (full proof §5.2, p. 29). The proof follows the same ideas as Theorem 1.2 but is significantly simpler — some steps are omitted — using $\Phi^T_{\mathrm{fin}} := \Phi^T_{\theta_3} \circ \Phi^{2T/3}_{\theta_2} \circ \Phi^{T/3}_{\theta_1}$. **(1) Disentangle:** Φ_{θ_1} on $[0, T/3]$ from Proposition 3.1, with $\|\theta_1\| \lesssim d\,N/T$. **(2) Cluster:** Φ_{θ_2} on $[T/3, 2T/3]$ from Proposition 2.1, applicable since the $\Phi^{T/3}(\mu_0^i)$ are now disjoint and single-hemisphere, driving each to within δ of a point $\delta_{x_0^i}$ with $T_\delta = O(\log 1/\delta)$, so $\|\theta_2\| \lesssim \log 1/\delta$. **(3) Match:** Φ_{θ_3} on $[2T/3, T]$ from Proposition 4.1 steers the ensemble of atoms x_0^i to the targets y^i, with $\|\theta_3\| \lesssim N/T$; since all measures are δ-close to $\delta_{x_0^i}$ the error is $e^{O(N\cdot T)}\delta$. All in all $\mathsf{W}_2(\mu^i(T), \delta_{y^i}) \leq \varepsilon$ with $\|\theta\|_{L^\infty((0,T);\Theta)} = O(d \cdot N/T + \log 1/\varepsilon)$.

Our formalization. open

The paper's proof (Geshkovski, Rigollet, Ruiz-Balet, §5.2) is the simpler Dirac-target specialization: disentangle by Proposition 3.1 ($\|\theta_1\| \lesssim dN/T$), cluster each now-disjoint single-hemisphere measure to within δ of a point by Proposition 2.1 ($T_\delta = O(\log 1/\delta)$, $\|\theta_2\| \lesssim \log 1/\delta$), then match the ensemble of cluster atoms x_0^i to the targets y^i by Proposition 4.1 ($\|\theta_3\| \lesssim N/T$); Lemma 5.4 is not needed. The end-to-end construction is validated numerically by experiment **E6**, while the negative control **E7** shows that a single *linear* continuity equation cannot achieve it — motivating the measure-dependent vector field. We give a type-correct Lean **statement** as a `sorry` (open): a shared missing direction and Dirac targets δ_{x^i} yield a single piecewise-constant θ with `W2 (measureFlow θ T (μ₀ i)) (Measure.dirac (x i)) ≤ ε` for every i. The $O(d\,N)$ switch count and the norm bound $O(dN/T + \log 1/\varepsilon)$ are documented in this blueprint prose, not encoded in the Lean type. We do not claim a proof.

```
/-- **Theorem 1.1** (Dirac targets). If the targets are point masses `δ_{x i}` and the
↪  inputs share
a missing direction, then for any horizon and tolerance a single piecewise-constant `θ`
↪  steers each
input to within `ε` of its target in `W₂`. Stub (`sorry`): status `math.open`. The proof
↪  also gives
`O(d N)` switches and the stated norm bound (see the blueprint). -/
theorem theorem_1_1 (hd : 3 ≤ d) {N : ℕ} (μ₀ : Fin N → Measure (Eucl d)) (x : Fin N →
↪  Eucl d)
    (T ε : ℝ) (hT : 0 < T) (hε : 0 < ε) (hmiss : SharedMissingDirection μ₀) :
    ∃ θ : Params d, ∀ i, W2 (measureFlow θ T (μ₀ i)) (Measure.dirac (x i)) ≤ ε := by
  sorry
```

Statements/MainResults.lean

6.5 Proposition 6.2 — generic O(1) disentanglement

From the original paper (Geshkovski, Rigollet, Ruiz-Balet, Proposition 6.2, p. 30). Let $d \geq 3$, $\pi = \mathrm{Unif}((\mathbb{S}^{d-1})^n)$, and sample $\mu_0^1, \ldots, \mu_0^N$ i.i.d. from

$$\mathcal{S}_n := \left\{ \frac{1}{n} \sum_{k=1}^{n} \delta_{x_k} : x_k \in \mathbb{S}^{d-1} \right\} \simeq (\mathbb{S}^{d-1})^n.$$

Consider (1.3)–(1.2) with $\mathbf{B} \equiv \beta I_d$ ($\beta \geq 0$), $\mathbf{V} \equiv I_d$, $\mathbf{W} \equiv 0$. Let x_*^i be the (a.s. existing) limit cluster point of $\mu^i(t)$. Then $\mathbb{P}[x_*^i \neq x_*^j] = 1$ for $i \neq j$.

Our formalization. informal

Reviewed on paper; not stated in Lean. The paper's argument (sketch of proof, p. 30) is a symmetry/measure-theoretic one: for π-almost every empirical input the dynamics with $\mathbf{B} \equiv \beta I_d$, $\mathbf{V} \equiv I_d$, $\mathbf{W} \equiv 0$ converge to a Dirac δ_X, defining a map $f : \mathcal{S}_n \to \mathbb{S}^{d-1}$. Because the flow and the law π are $\mathbb{O}(d)$-equivariant, the law $\nu = f_{\#}\pi$ of the limit is rotation-invariant, hence non-atomic — an atom at x_0 of mass p would force an atom of mass p at every $\mathsf{R}x_0$, contradicting $\nu(\mathbb{S}^{d-1}) = 1$ once enough distinct rotations are chosen. Two independent inputs then give independent limits with this common non-atomic law, so $\mathbb{P}[x_*^i = x_*^j] = \int \nu(\{x\})\, \mathrm{d}\nu(x) = 0$. We do not state this in Lean: it would require Haar measure on $\mathbb{O}(d)$, the equivariant ω-limit map f, and non-atomicity infrastructure outside our current axiom layer. The result is recorded as the heuristic that the generic switch complexity is $O(1)$ rather than the $O(d\,N)$ of Theorem 1.1.